

User Manual

for S32 RM Driver

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Chapter 1

Revision History

Revision	Date	Author	Description
1.0	30.06.2023	NXP RTD Team	S32 Real-Time Drivers AUTOSAR 4.4 Version 4.0.2 Release

Chapter 2

Introduction

- [Supported Derivatives](#)
- [Overview](#)
- [About This Manual](#)
- [Acronyms and Definitions](#)
- [Reference List](#)

This User Manual describes NXP Semiconductor Resource Manager for S32. Resource Manager driver configuration parameters and deviations from the specification are described in Driver chapter of this document.

2.1 Supported Derivatives

The software described in this document is intended to be used with the following microcontroller devices of NXP Semiconductors:

- s32g274a_bga525
- s32g254a_bga525
- s32g233a_bga525
- s32g234m_bga525
- s32g378a_bga525
- s32g379a_bga525
- s32g398a_bga525
- s32g399a_bga525
- s32g338m_bga525
- s32g339m_bga525
- s32g358a_bga525
- s32g359a_bga525
- s32r45_bga780

All of the above microcontroller devices are collectively named as S32.

2.2 Overview

AUTOSAR (AUTomotive Open System ARchitecture) is an industry partnership working to establish standards for software interfaces and software modules for automobile electronic control systems.

AUTOSAR:

- paves the way for innovative electronic systems that further improve performance, safety and environmental friendliness.
- is a strong global partnership that creates one common standard: "Cooperate on standards, compete on implementation".
- is a key enabling technology to manage the growing electrics/electronics complexity. It aims to be prepared for the upcoming technologies and to improve cost-efficiency without making any compromise with respect to quality.
- facilitates the exchange and update of software and hardware over the service life of the vehicle.

2.3 About This Manual

This Technical Reference employs the following typographical conventions:

- **Boldface** style: Used for important terms, notes and warnings.
- *Italic* style: Used for code snippets in the text. Note that C language modifiers such "const" or "volatile" are sometimes omitted to improve readability of the presented code.

Notes and warnings are shown as below:

Note

This is a note.

Warning

This is a warning

2.4 Acronyms and Definitions

Term	Definition
API	Application Programming Interface
ASM	Assembler
BSMI	Basic Software Make file Interface
C/CPP	C and C++ Source Code
DEM	Diagnostic Event Manager
DET	Development Error Tracer
ECU	Electronic Control Unit
LSB	Least Significant Bit
MCU	Micro Controller Unit
MIDE	Multi Integrated Development Environment
MOU	Memory Protection Unit
MRC	Memory Region Controller
MSB	Most Significant Bit
MSCM	Miscellaneous System Control Module
MPU_M7	Arm Cortex M7 Memory Protection Unit
N/A	Not Applicable
RAM	Random Access Memory
RM	Resource Manager
SEMA42	Semaphores 2
SIU	Systems Integration Unit
SWS	Software Specification
XML	Extensible Markup Language
XRDC	Extended Resource Domain Controller

2.5 Reference List

#	Title	Version
1	S32G2 Reference Manual	Rev. 7, February 2023
2	S32G3 Reference Manual	Rev. 3, Feb 2023
3	S32G2: Mask Set Errata for Mask 0P77B	Rev. 3.0, 4/2023
4	S32G2: Mask Set Errata for Mask 1P77B	Rev. 1.0, 4/2023
5	S32G3: Mask Set Errata for Mask 1P72B	Rev. 2.1, 4/2023
6	S32G2 Data Sheet	Rev. 6, Dec 2022
7	S32G3 Data Sheet	Rev. 2, RC, Feb 2023
8	VR5510 Data Sheet	Rev. 5, 6 April 2022
9	S32G2 Safety Manual	Rev. 1.1, March 2021
10	S32G3 Safety Manual	Rev. 2.1, Feb 2023
11	ARMv7-M Architecture Reference Manual (for Mpu M7)	ARM DDI 0403E.b
12	S32R45 Reference Manual	Rev. 3, 12/2021
13	S32R45 Data Sheet	Rev. 3, RC, 11/2022

Introduction

#	Title	Version
14	S32R45 Mask Set Errata for Mask P57D	Rev. 2.0
15	S32R45 Safety Manual	Rev. 2, Dec 2021

Chapter 3

Driver

- [Requirements](#)
- [Driver Design Summary](#)
- [Hardware Resources](#)
- [Deviations from Requirements](#)
- [Driver Limitations](#)
- [Driver usage and configuration tips](#)
- [Runtime errors](#)
- [Symbolic Names Disclaimer](#)

3.1 Requirements

Resource Manager is a Complex Device Driver (CDD), so there are no AUTOSAR requirements regarding this module.

It has vendor-specific requirements and implementation.

3.2 Driver Design Summary

The RM module provides a way to initialize and control the resource domains allocation and memory protection on the chip with the supported peripherals.

3.3 Hardware Resources

#	Hardware IP	Description
1	MPU_M7	Arm Cortex M7 Memory Protection Unit
2	XRDC	Extended Resource Domain Controller
3	Sema42	Semaphores 2
4	Mscm	Miscellaneous System Control Module

The RM driver uses MPU_M7, XRDC, Sema42 and Mscm hardware IP.

3.4 Deviations from Requirements

Since this is a CDD Module, there are no AUTOSAR requirements for the functionality.

3.5 Driver Limitations

- MPU_M7:
 - Multi-variant is not supported.
- XRDC:
 - For XRDC on **S32G3 Rev 1.1**: There is only one MDA_Wx_y_DFMTz register per PFE MDAC. The users have use the DID bypass mode to isolate the PFE Host Interfaces. The workaround is described in the chapter [XRDC PFE Host Interfaces Isolation](#).
- RM:
 - Cache limitation: due to MPU does not handle cache coherency, Rm_Init might get affected from changing region attribute (from cacheable to non-cacheable). Cache sync should be required in this case. In this case please follow step:
 Step 1: Cache clean (Invalidate and flush)
 Step 2: Cache disable
 Step 3: Rm_Init/ Mpu_M7_Ip_Init
 Step 4: Cache_enable
 - Interrupt limitation: Changing attribute of region dynamically might get problem with interrupt, disable/enable interrupt might be required.

3.6 Driver usage and configuration tips

3.6.1 HLD usage

Prior usage of the RM CDD in an application, the configuration files must be generated with the configurator.

3.6.1.1 API controls

Initialization of all hardware resources is done using [Rm_Init](#). Considering that the function is configuring domains and access to resources it is recommended that it is called before other bus masters are active, or that they are not accessing those resources.

The rest of the API controls directly the allocated hardware resources. Check the references below for more information:

- [Rm_XrdcSetProcessID](#)
- [Rm_XrdcGetDomainID](#)
- [Rm_XrdcGetDomainIDErrorStatus](#)
- [Rm_SemaphoreGetStatus](#)
- [Rm_SemaphoreLockGate](#)
- [Rm_SemaphoreUnlockGate](#)
- [Rm_SemaphoreResetGate](#)
- [Rm_SemaphoreResetAllGates](#)
- [Rm_Mpu_M7_SetRegionConfig](#)
- [Rm_Mpu_M7_EnableRegion](#)
- [Rm_Mpu_M7_SetAccessRight](#)
- [Rm_Mpu_M7_GetErrorDetails](#)
- [Rm_Mpu_M7_MulticoreInit](#)

3.6.2 LLD usage

Prior usage of the LLDs from RM CDD in an application, the configuration files must be generated with the S32CT configurator.

3.6.2.1 API controls

For more information on the LLD API check the references below:

- [Mpu M7 IPV Driver](#)
 - [Mpu_M7_Ip_Init](#)
 - [Mpu_M7_Ip_MulticoreInit](#)
 - [Mpu_M7_Ip_SetRegionConfig](#)
 - [Mpu_M7_Ip_Deinit](#)
 - [Mpu_M7_Ip_EnableRegion](#)
 - [Mpu_M7_Ip_SetAccessRight](#)

- Mpu_M7_Ip_GetErrorDetails

- [XRDC IPV Driver](#)

- Xrdc_Ip_Init
- Xrdc_Ip_GetDomainID
- Xrdc_Ip_SetProcessID
- Xrdc_Ip_GetDomainIDErrorStatus

- [Sema42 IPV Driver](#)

- Sema42_Ip_Init
- Sema42_Ip_LockGate
- Sema42_Ip_UnlockGate
- Sema42_Ip_ResetGate
- Sema42_Ip_GetGateLocker
- Sema42_Ip_IsResetGateStateIdle
- Sema42_Ip_GetResetGateBusMaster
- Sema42_Ip_GetResetGateIndex

- [Mscm IPV Driver](#)

- Mscm_Ip_Init

3.6.3 Interface configuration

User can choose enable / disable checkbox to use the corresponding Module

General XRDC Domain Assignment XRDC Memory Config XRDC Peripheral Config MPU M7 Sema42 Published Information

Post Build Variant Used ☐ ▼

Config Variant VariantPreCompile ▼

▼ Rm General Configuration

Name RmGeneral

Rm_VersionInfoApi ☒ ▼ Rm Development Error Detect ☒ ▼

Rm Enable User Mode Support ☒ ▼ Rm Enable MultiCore Support ☐ ▼

RM Enable XRDC Support ☒ ▼ RM Enable Sema42 Support ☐ ▼

RM Enable MPU M7 Support ☒ ▼

▼ RmConfigSet

Name RmConfigSet

▼ Xrdc_Configuration

Name Xrdc_Configuration

Xrdc Development Error Detection ☒ ▼ Xrdc Registers Lock ☐ ▼

Xrdc PID Registers Lock XRDC_PID_UNLOCKED ▼

Figure 3.1 RM general config

3.6.4 Multicore initialization

RM is multicore type 3 but MPU can support type 2 multicore following autosar specifications. Because of that it require separated initialization for MPU and the rest of IP driver, If multicore is enable, the following init step should be required:

Core 0	Core 1	
step1: Rm_Init() (for master core only)	No initialization for Rm_Init()	type 3 multicore
step2: Do nothing(Mpu M7 Multicore was initialized in RM_Init())	Rm_Mpu_M7_Multicore_Init()	type 2 multicore

3.6.5 XRDC configuration

3.6.5.1 XRDC registers lock

When **XRDC Registers Lock** is enable, all Xrdc registers which have LK bit(s) except XRDC_PID m registers will be locked (read-only) until the next reset.

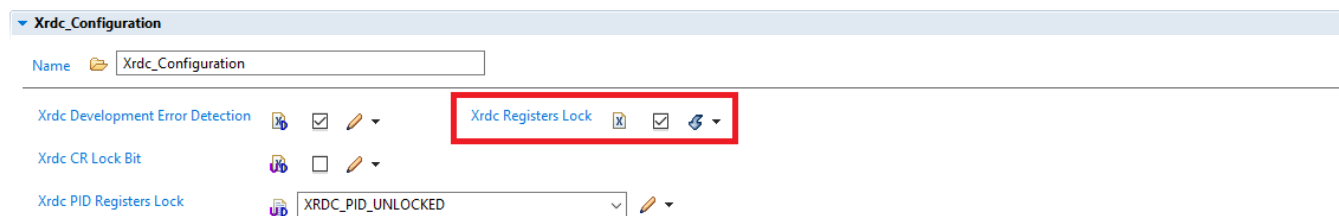


Figure 3.2 XRDC Registers lock

For XRDC PID Registers Lock , Tresos configuration provides users with 3 available options which are listed in figure and table below

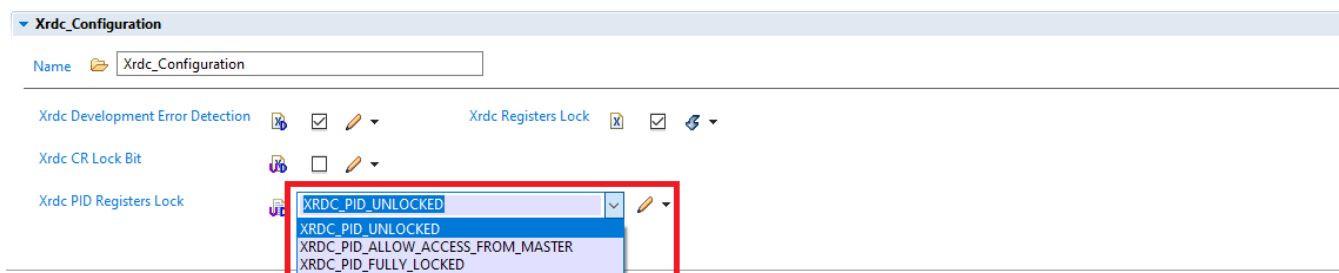


Figure 3.3 XRDC PID Registers lock options

Option	XRDC_PID m registers status
XRDC_PID_UNLOCKED	Register can be written to by any secure privileged write
XRDC_PID_ALLOW_ACCESS_FROM_MASTER	Register can only be written by a secure privileged write from bus master m
XRDC_PID_FULLY_LOCKED	Register locked (read-only) until the next reset

Note:

-If **XRDC Registers Lock** is enable, XRDC module cannot be disable during runtime. -**XRDC PID Registers Lock** will be aborted unless **XRDC Registers Lock** is enable.

3.6.5.2 XRDC Domain Assignment

Users can create a new domain assignment by click on Add button in configuration field.

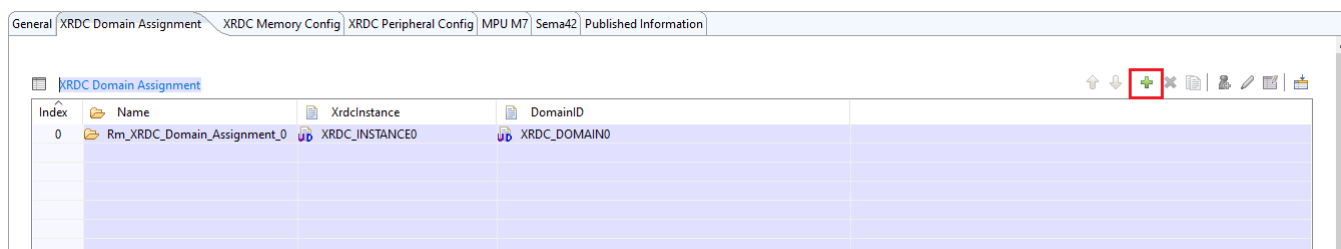


Figure 3.4 XRDC Domain Assignment Configuration

3.6.5.2.1 XRDC Master Domain Assignment

Users can add new elements to choose which masters will be assigned to the corresponding domain ID and configure their assignment attributes.

Rm_XRDC_Domain_Assignment

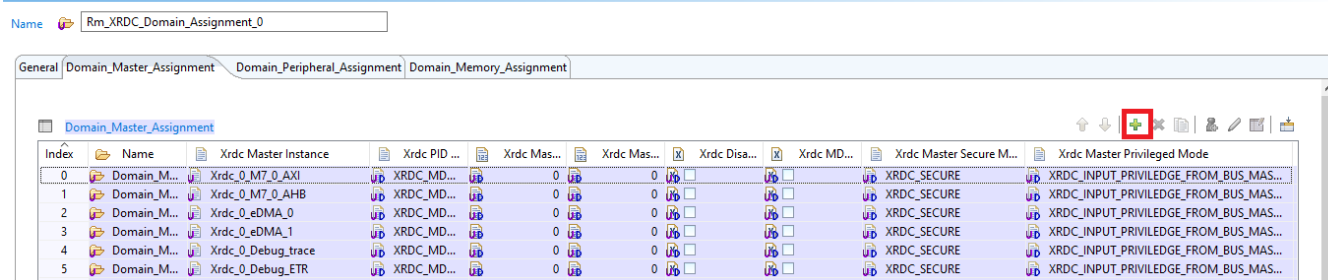


Figure 3.5 XRDC Domain Master Assignment Configuration

For ProcessID evaluation, Tresos configuration provides users with 3 available options which are listed in figure and table below

Rm_XRDC_Domain_Assignment

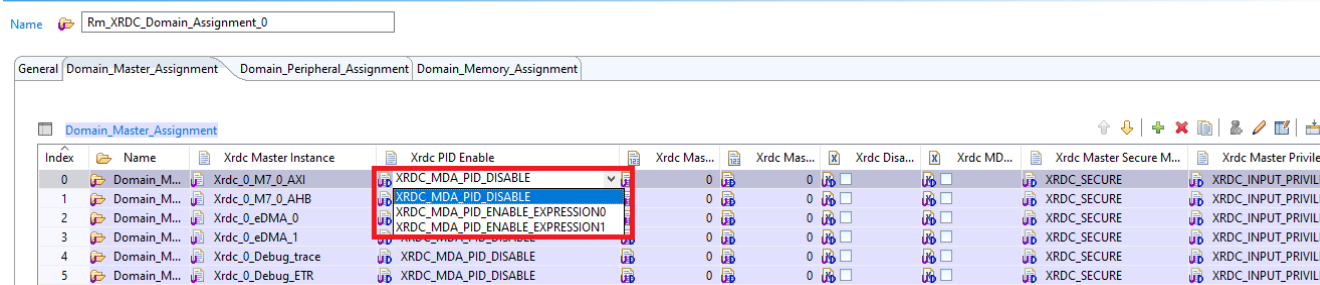


Figure 3.6 ProcessID options

Option	PID evaluation
XRDC_MDA_PID_DISABLE	No process identifier is included in the domain hit evaluation
XRDC_MDA_PID_ENABLE_EXPRESSION0	Domain assignment hit when $((PID \& \sim PIDM) == (PI \leftarrow Dn[PID] \& \sim PIDM))$
XRDC_MDA_PID_ENABLE_EXPRESSION1	Domain assignment hit when $\sim((PID \& \sim PIDM) == (PI \leftarrow Dn[PID] \& \sim PIDM))$

3.6.5.2.2 XRDC Peripheral Domain Assignment

Users can add new elements to configure domain access policy for a specific peripheral slot.

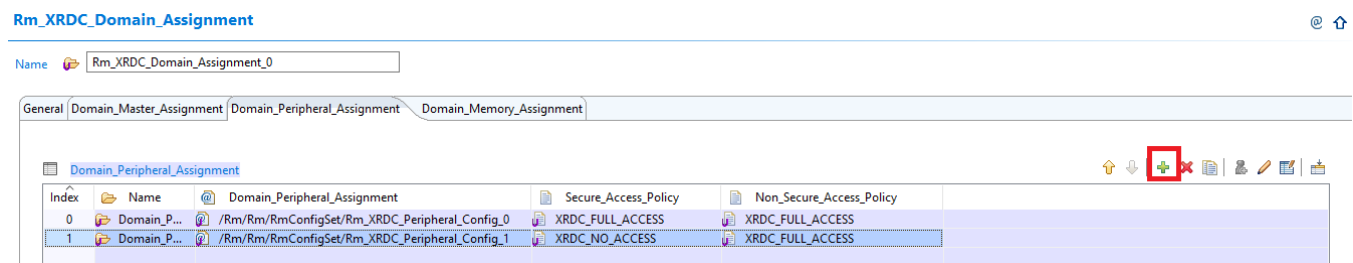


Figure 3.7 XRDC Domain Peripheral Assignment

Selections of **DomainPeripheralAssignment** field refer to [XRDC Peripheral Configuration](#)

Combination of **SecurityAccessPolicy** and **NonSecurityAccessPolicy** field defines final access policy for the peripheral slot (found in *Access policy look up table* shown below)

3.6.5.2.3 XRDC Memory Domain Assignment

Users can add new elements to configure domain access policy for a specific memory region.

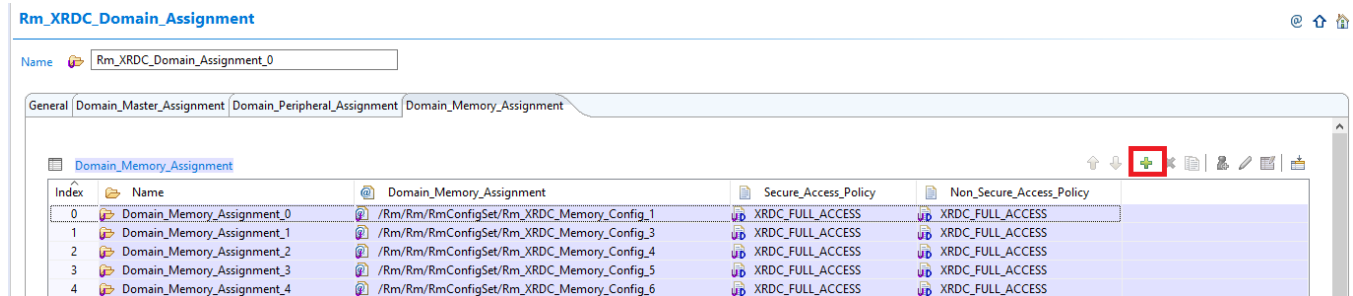


Figure 3.8 XRDC Domain Memory Assignment

Selections of **DomainMemoryAssignment** field refer to [XRDC Memory Configuration](#)

Combination of **SecurityAccessPolicy** and **NonSecurityAccessPolicy** field defines final access policy for the memory region (found in *Access policy look up table* shown below)

Configuration field		Allowable access			
SecureAccessPolicy	NonSecureAccessPolicy	SecurePriv	SecureUser	NonSecurePriv	NonSecureUser
XRDC_NO_ACCESS	X	none	none	none	none
XRDC_SECURE_READ_ONLY	X	r	r	none	none
XRDC_SECURE_SUPERVISOR_ACCESS_ONLY	X	r,w	none	none	none
XRDC_SECURE_FULL_ACCESS	XRDC_NON_SECURE_NO_ACCESS	r,w	r,w	none	none
XRDC_SECURE_FULL_ACCESS	XRDC_NON_SECURE_SUPERVISOR_READ_ONLY	r,w	r,w	r	none
XRDC_SECURE_FULL_ACCESS	XRDC_NON_SECURE_READ_ONLY	r,w	r,w	r	r
XRDC_SECURE_FULL_ACCESS	XRDC_NON_SECURE_SUPERVISOR_ACCESS_ONLY	r,w	r,w	r,w	none
XRDC_SECURE_FULL_ACCESS	XRDC_NON_SECURE_FULL_ACCESS	r,w	r,w	r,w	r,w

Figure 3.9 Access policy look up table

3.6.5.3 XRDC Peripheral Configuration

Users can add new elements to configure attributes of peripheral slots

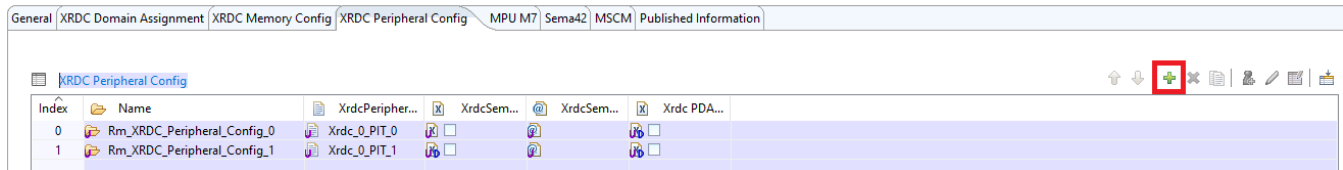


Figure 3.10 XRDC Domain Peripheral Configuration

Note:

- XrdcSema4Number** will be aborted unless **XrdcSema4Enable** is checked.
- XrdcSema4Number** indicates the index of hardware semaphore gates. However, these gates can be mapped to different logical channels mentioned in [Sema42 logic channel configuration](#)

3.6.5.4 XRDC Memory Configuration

Users can add new elements to configure attributes of memory regions.

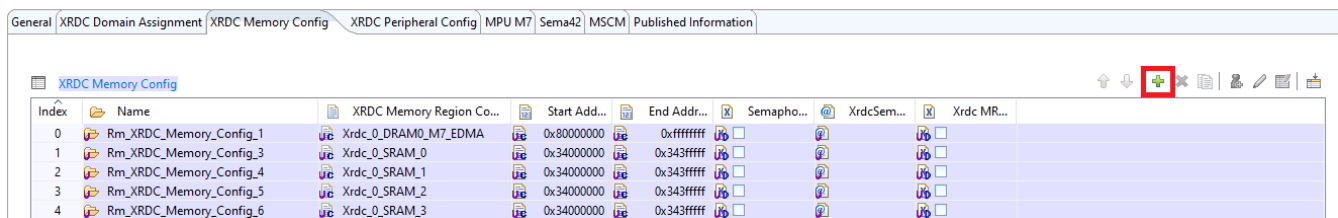


Figure 3.11 XRDC Domain Memory Configuration

Note:

- XrdcSema4Number** will be aborted unless **XrdcSema4Enable** is checked.
- XrdcSema4Number** indicates the index of hardware semaphore gates. However, these gates can be mapped to different logical channels mentioned in [Sema42 logic channel configuration](#)
- Each memory region need to be protected by an appropriate MRC instance, details are mentioned in the [Reference manual](#)

3.6.5.5 XRDC PFE Host Interfaces Isolation

3.6.5.5.1 Background

HIF – HOST INTERFACE

- 4x DMA Tx Channels
- 4x DMA Rx Channels
- 1x HIF_NCP Channel
- Separate Rx/Tx BD chains with per channel interrupts
- 4x HIF Interfaces
 - HIF_M1 – BD Fetch
 - HIF_M2 – BD WriteBack
 - HIF_M3 – Packet Write
 - HIF_M4 – Packet Read

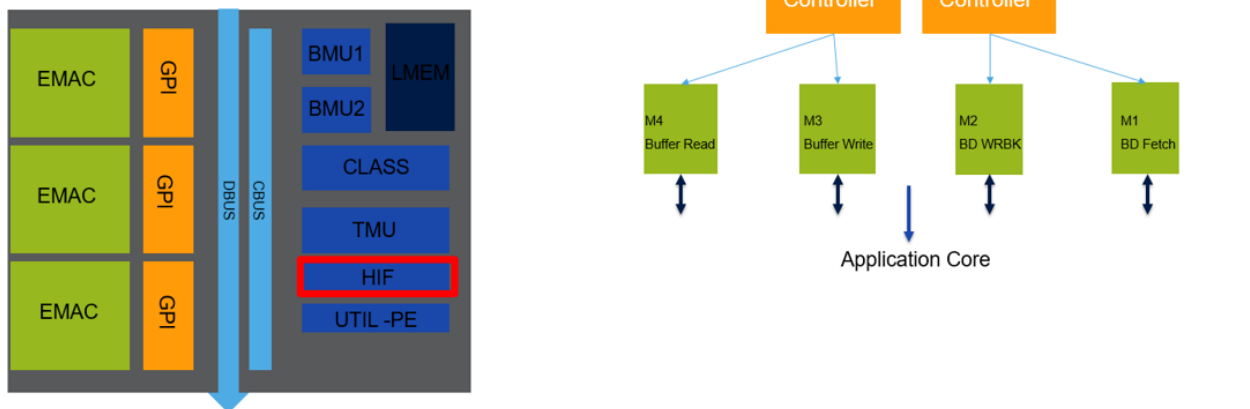


Figure 3.12 PFE - HIF (Host interface)

- The PFE Host Interface Isolation Feature can work by using XRDC
- Each host interface has 4 channels and we can assign each of them to a different XRDC domain
- The PFE's user bits will output an LPID and the MDAC would map each LPID to a unique DID

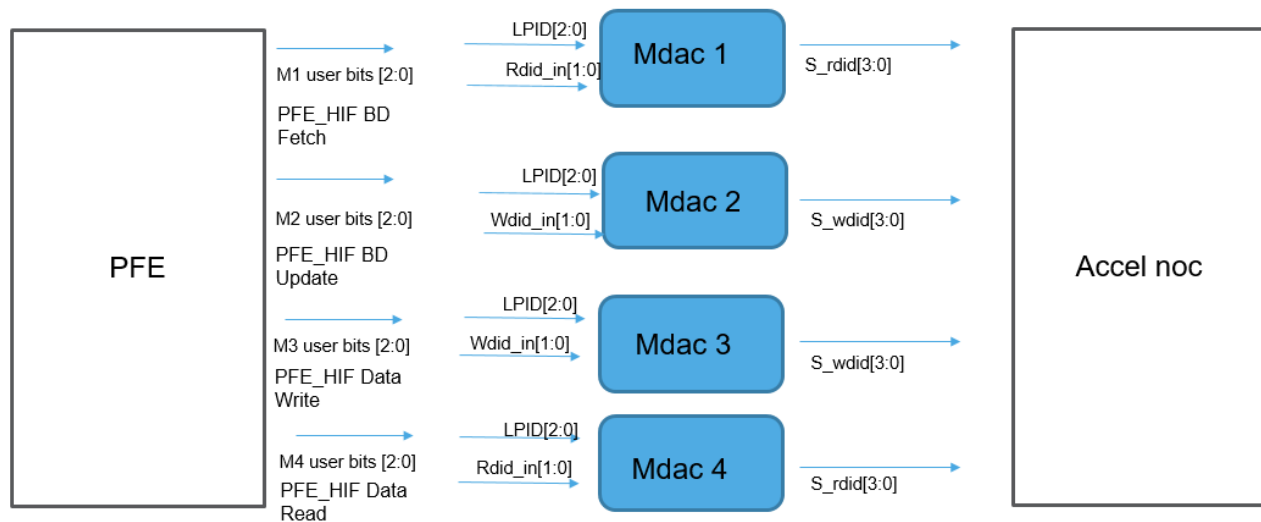


Figure 3.13 Current limitation using LPID

- But the MDACs have been configured in such a way that this would not be possible (because they only have a single DFMT per MDAC meaning they can only be part of 1 domain)

3.6.5.5.2 Issue

- In order for a port to support multiple DIDs, the MDAC must have multiple **MDA_Wx_y_DFMTz[DIDB]** registers
- For the case of the S32G3 there should be 4 registers per PFE MDAC, however there is only 1
- This means that we cannot have different DIDs per port and therefore we are no better off than we were on S32G2
- Each PFE MDAC has only one **MDA_Wx_y_DFMTz[DIDB]** register, meaning that all 4 channels can only be part of 1 domain
- In short, LPID mode cannot be used at this moment because there are not enough descriptors

3.6.5.5.3 Workaround

- Use the DID bypass mode **MDA_Wx_y_DFMTz[DIDB]** to allow variability in the DID
- In this mode, the incoming **DID_in** will be used as the output DID domain

Domain_Master_Assignment

Name  Domain_Master_Assignment_6

General

XrdcMasterInstance	 PFE_HIF0  
XrdcPIDEnable	 XRDC_MDA_PID_DISABLE 
XrdcMasterPID (0 -> 63)	 0
XrdcMasterPIDMask (0 -> 63)	 0
Xrdc Disable DID Bypass	 ☒  
Xrdc Master Secure Mode	 XRDC_SECURE  
XrdcCoreMasterDescriptor	 XRDC_CORE_DESCRIPTOR_0 
Xrdc Master Privileged Mode	 XRDC_INPUT_PRIVILEGE_FROM_BUS_MASTER  

Figure 3.14 XRDC DID bypass mode

3.6.6 Sema42 logic channel configuration

Users can use logic channel instead of hardware instance and hardware channel. The logic channel is stand for both hardware instance and hardware channel which is configured in configuration application.

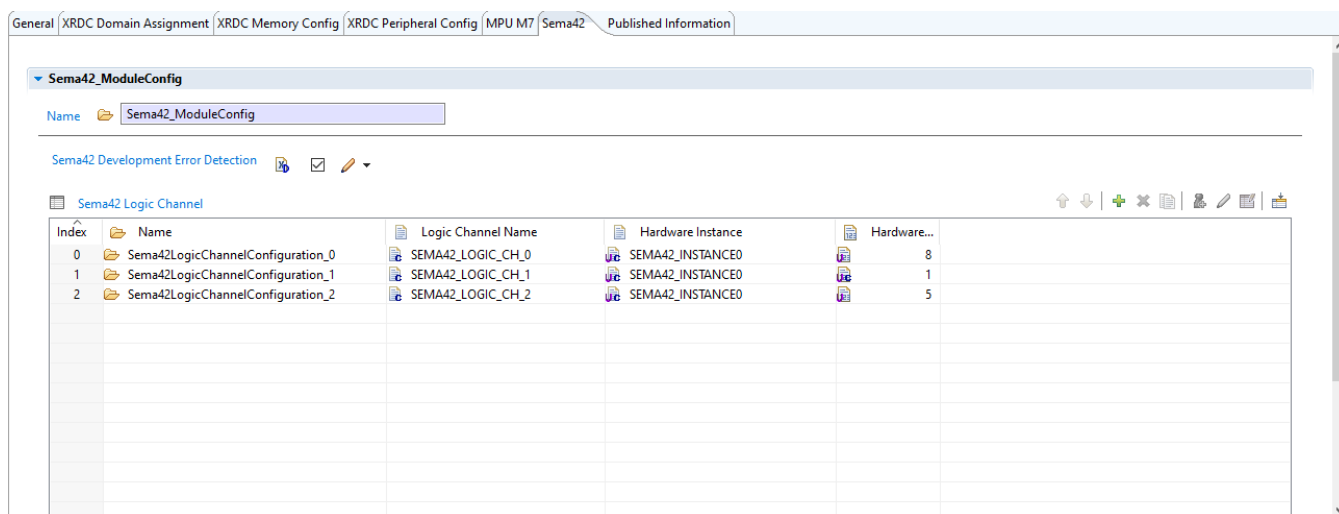


Figure 3.15 Sema42 LogicChannel Configuration

Example: In the picture, there are 3 logic channel:

- Logic channel 1 is equal to Sema42 instance 0 gate 8.
- Logic channel 2 is equal to Sema42 instance 0 gate 1.
- Logic channel 3 is equal to Sema42 instance 0 gate 5.

Note: In Rm_Semaphore API, we should use the name which is defined in column **Logic Channel Name** in configuration application with the prefix "RM_" (Because it is a macro so all characters are uppercase). We can find out them in file "CDD_Rm_Ipw_Cfg.h" with identify "RM_<Logic Channel Name>". As in picture, if users want to invoke Sema42 instance 0 gate 5, the argument they must pass into Rm_Semaphore API is RM_SEMA42_LOGIC_CH_2.

3.6.7 MPU M7 configuration

The MPU divides the memory map into a number of regions, and defines the location, size, access permissions, and memory attributes of each region. Details refer to "ARMv7-M Architecture Reference Manual" documentation of the ARM. (Chapter B3.5 Protected Memory System Architecture, PMSAv7)

A Mpu M7 Region Configuration Interface as shown below:

RegionConfig

Name

General

Region Number (dynamic range)	1
Start Address (0x0 -> 0xffffffff)	0x0
End Address (dynamic range)	0x8000
Region Size	0x8001
Memory type	PU_M7_MEM_STRONG_ORDER
Access Rights	M7_PRIV_RWX_UNPRIV_NONE
Outer Cache Policy	17_CACHE_POLICY_NO_CACHE
Inner Cache Policy	17_CACHE_POLICY_NO_CACHE
Subregion Mask (0 -> 255)	0
Shareable	☒

Default

Figure 3.16 Mpu M7 Region Configuration Interface.

The **Memory Type** node allows to select the attribute for the memory region shown in the picture below.

Mpu M7 Region Attribute. (Source : "ARMv7-M Architecture Reference Manual" of the ARM.)

Region Memory Type in the Tressos interface

Table B3-13 TEX, C, B, and S Encoding

TEX	C	B	Memory type	Description, or Normal region Cacheability	Shareable?
000	0	0	Strongly-ordered	Strongly ordered	Shareable
000	0	1	Device	Shared device	Shareable
000	1	0	Normal	Outer and inner Write-Through, no write allocate	S bit ^a
000	1	1	Normal	Outer and inner write-back, no write allocate	S bit ^a
001	0	0	Normal	Outer and inner Non-cacheable	S bit ^a
001	0	1	Reserved	Reserved	Reserved
001	1	0	IMPLEMENTATION DEFINED	IMPLEMENTATION DEFINED	IMPLEMENTATION DEFINED
001	1	1	Normal	Outer and inner write-back; write and read allocate	S bit ^a
010	0	0	Device	Non-shared device	Non-shareable
010	0	1	Reserved	Reserved	Reserved
010	1	X	Reserved	Reserved	Reserved
011	X	X	Reserved	Reserved	Reserved
1BB	A	A	Normal	Cached memory, with AA and BB indicating the inner and outer Cacheability rules that must be exported on the bus. See Table B3-14 on page B3-642 for the Cacheability policy encoding. BB = Outer policy, AA = Inner policy.	S bit ^a

a. Shareable if the S bit is set to 1, Non-shareable if the S bit is set to 0

Figure 3.17 Mpu M7 Region Memory Type.

Note:

- Only if the configuration **Memory type** is "MPU_M7_MEM_NORMAL_CACHEABLE", "Outer Cache Policy"/"Inner Cache Policy" makes sense.
- When configuration **Memory type** is "MPU_M7_MEM_STRONG_ORDER", "MPU_M7_MEM_DEVICE_SHARED", means region is always Shareable, the **Shareable** node value has no meaning.
- When configuration **Memory type** is "MPU_M7_MEM_DEVICE_NOSHARE", means region is always Non-Shareable, the **Shareable** node value has no meaning.

3.6.8 Errata for MPU M7:

[Core] Speculative accesses might be performed to memory unmapped in MPU. Cortex-M7 can incorrectly perform speculative accesses to unmapped regions. Instead of leaving memory unmapped, the RM driver recommends the user to use MPU region 0 to cover all memory (0x0-0xffffffff) and make this region execute-never and inaccessible. In tressos interface, default region 0 should be configured as shown below:

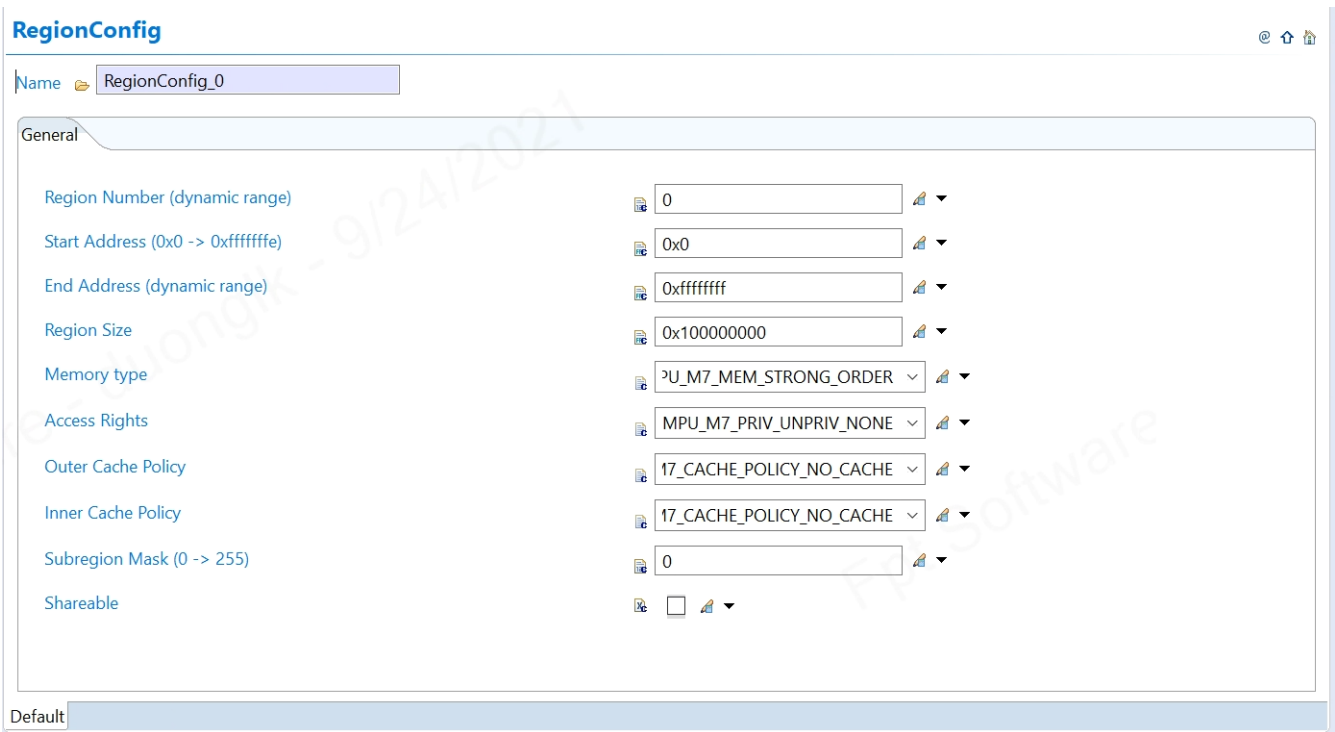


Figure 3.18 Default region 0

Note that the MPU supports addressing hitting in multiple regions with the highest numbered region taking priority. Therefore, use of MPU region 0 in this way does not affect the existing organization and use of MPU regions

3.6.9 MSCM configuration

Mscm configuration support users configure Mscm by instance. **MSCM Configuration** table contains the instance that will be configured.

The screenshot shows the 'Mscm_Configuration' window. The 'Name' field is set to 'Mscm_Configuration'. Below the field, there is a table titled 'MSCM Configuration' with the following data:

Index	Name	Interrupt Name	GIC500	M7_0_CO...	M7_1_CO...	M7_2_CO...	NOT_USED
0	MscmConfig_0	MSCM_PIT_0	☒	☒	☒	☒	☒
1	MscmConfig_1	MSCM_PIT_1	☒	☒	☒	☒	☒

Figure 3.19 MSCM Instance Configuration

This application can enable and disable individual interrupts for each core in a multi-core system.

MscmConfig

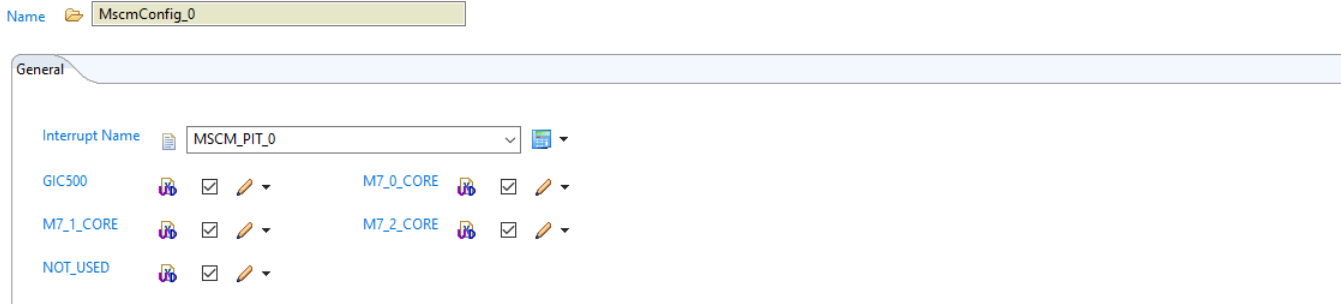


Figure 3.20 MSCM Configuration

3.7 Runtime errors

The driver does not trigger any DEM runtime errors, but triggers the runtime DET errors listed in the table below:

Function	Error Code	Condition triggering the error
Rm_Init()	RM_E_UNINIT_U8	API service called without module initialization.
Rm_Init()	RM_E_ALREADY_INITIALIZED_U8	Module is already initialized
Rm_Init()	RM_E_INIT_FAILED_U8	In precompile mode, specifies that a non-null parameter was used for init. In other cases, a null pointer was used for initialization
Rm_XrdcGetDomainID()	RM_E_XRDC_UNKNOWN_DOMAINID_U8	The domain ID currently retrieved is not in the supported range.
Rm_SemaphoreGetStatus(), Rm_SemaphoreLockGate(), Rm_SemaphoreUnlockGate(), Rm_SemaphoreResetGate()	RM_E_INVALID_SEMA4_CHANNEL_U8	Semaphore channel is invalid
Rm_SemaphoreLockGate()	RM_E_ALREADY_LOCKED_U8	Semaphore is already locked
Rm_SemaphoreLockGate(), Rm_SemaphoreUnlockGate()	RM_E_LOCKED_BY_OTHER_CORE_U8	Semaphore is locked by another core
Rm_SemaphoreLockGate()	RM_E_ALREADY_UNLOCKED_U8	Semaphore is already unlocked
Rm_SemaphoreLockGate(), Rm_SemaphoreUnlockGate()	RM_E_SEMAPHORE_UNKNOWN_DOMAINID_U8	The domain ID currently retrieved (get from Sema42 hardware) is not in the supported range.
Rm_GetVersionInfo()	RM_E_PARAM_POINTER	API is called with invalid pointer parameter.

3.8 Symbolic Names Disclaimer

All containers having `symbolicNameValue` set to `TRUE` in the AUTOSAR schema will generate defines like:

```
#define <Mip>Conf_<Container_ShortName>_<Container_ID>
```

For this reason it is forbidden to duplicate the names of such containers across the RTD configurations or to use names that may trigger other compile issues (e.g. match existing `#ifdefs` arguments).

Chapter 4

Tresos Configuration Plug-in

This chapter describes the Tresos configuration plug-in for the driver. All the parameters are described below.

- Module [Rm](#)
 - Container [RmGeneral](#)
 - * Parameter [Rm_VersionInfoApi](#)
 - * Parameter [RmDevErrorDetect](#)
 - * Parameter [RmEnableUserModeSupport](#)
 - * Parameter [RmEnableMultiCore](#)
 - * Parameter [RmEnableXRDCSupport](#)
 - * Parameter [RmEnableSema42Support](#)
 - * Parameter [RmMpuM7Configurable](#)
 - * Parameter [RmEnableMscmSupport](#)
 - Container [RmConfigSet](#)
 - * Container [Xrdc_Configuration](#)
 - Parameter [XrdcDevErrorDetect](#)
 - Parameter [XrdcRegistersLock](#)
 - Parameter [XrdcCRLockBit](#)
 - Parameter [XrdcPIDRegisterLock](#)
 - * Container [Rm_XRDC_Domain_Assignment](#)
 - Parameter [DomainID](#)
 - Container [Domain_Master_Assignment](#)
 - Parameter [XrdcMasterInstance](#)
 - Parameter [XrdcPIDEnable](#)
 - Parameter [XrdcMasterPID](#)
 - Parameter [XrdcMasterPIDMask](#)
 - Parameter [XrdcDIDBDisable](#)
 - Parameter [XrdcMDADFMTLockBit](#)
 - Parameter [XrdcMasterMode](#)
 - Parameter [XrdcMasterPriviledgeMode](#)
 - Container [Domain_Peripheral_Assignment](#)
 - Parameter [Secure_Access_Policy](#)
 - Parameter [Non_Secure_Access_Policy](#)

- Reference [Domain_Peripheral_Assignment](#)
- Container [Domain_Memory_Assignment](#)
- Parameter [Secure_Access_Policy](#)
- Parameter [Non_Secure_Access_Policy](#)
- Reference [Domain_Memory_Assignment](#)
- * Container [Rm_XRDC_Memory_Config](#)
 - Parameter [XrdcMrcInstance](#)
 - Parameter [XrdcStartAddress](#)
 - Parameter [XrdcEndAddress](#)
 - Parameter [XrdcSema4Enable](#)
 - Parameter [XrdcMRGDLockBit](#)
 - Reference [XrdcSema42LogicChannel](#)
- * Container [Rm_XRDC_Peripheral_Config](#)
 - Parameter [XrdcPeripheralSlot](#)
 - Parameter [XrdcSema4Enable](#)
 - Parameter [XrdcPDACLockBit](#)
 - Reference [XrdcSema42LogicChannel](#)
- * Container [MPU_M7_Configuration](#)
 - Parameter [MpuDevErrorDetect](#)
 - Container [MPU_M7_ModuleConfig](#)
 - Parameter [DefaultMapEnable](#)
 - Parameter [RunInHFNMIEEnable](#)
 - Parameter [MemManageInterruptEnable](#)
 - Parameter [PhysicalCoreID](#)
 - Container [RegionConfig](#)
 - Parameter [RegionNumber](#)
 - Parameter [StartAddress](#)
 - Parameter [EndAddress](#)
 - Parameter [RegionSize](#)
 - Parameter [MemoryType](#)
 - Parameter [AccessRights](#)
 - Parameter [OuterCachePolicy](#)
 - Parameter [InnerCachePolicy](#)
 - Parameter [SubregionMask](#)
 - Parameter [Shareable](#)

- * Container [Sema42_ModuleConfig](#)
 - Parameter [Sema42DevErrorDetect](#)
 - Parameter [Sema42ResetAllGates](#)
 - Container [Sema42LogicChannelConfiguration](#)
 - Parameter [Sema42LogicChannel_LogicName](#)
 - Parameter [Sema42HardwareInstance](#)
 - Parameter [Sema42HardwareChannel](#)
- * Container [Mscm_Configuration](#)
 - Parameter [MscmDevErrorDetect](#)
 - Container [MscmConfig](#)
 - Parameter [IsrName](#)
 - Parameter [IsrTargetCore0](#)
 - Parameter [IsrTargetCore1](#)
 - Parameter [IsrTargetCore2](#)
 - Parameter [IsrTargetCore3](#)
 - Parameter [IsrTargetCore4](#)
- Container [CommonPublishedInformation](#)
 - * Parameter [ArReleaseMajorVersion](#)
 - * Parameter [ArReleaseMinorVersion](#)
 - * Parameter [ArReleaseRevisionVersion](#)
 - * Parameter [ModuleId](#)
 - * Parameter [SwMajorVersion](#)
 - * Parameter [SwMinorVersion](#)
 - * Parameter [SwPatchVersion](#)
 - * Parameter [VendorApiInfix](#)
 - * Parameter [VendorId](#)

4.1 Module Rm

Vendor specific: Configuration of the Rm (Resource Manager) module.

Included containers:

- [RmGeneral](#)
- [RmConfigSet](#)
- [CommonPublishedInformation](#)

Property	Value
type	ECUC-MODULE-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantSupport	true
supportedConfigVariants	VARIANT-POST-BUILD, VARIANT-PRE-COMPILE

4.2 Container RmGeneral

Vendor specific: Configuration of general Rm parameters.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.3 Parameter Rm_VersionInfoApi

Vendor specific: Enables/Disables the get version info API function

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.4 Parameter RmDevErrorDetect

Vendor specific:

Switches the Development Error Detection and Notification on or off.

true: Enabled.

false: Disabled.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
default Value	true

4.5 Parameter RmEnableUserModeSupport

When this parameter is enabled, the RM module will adapt to run from User Mode, with the following measures:

b) using 'call trusted function' stubs for all internal function calls that access registers requiring supervisor mode.

for more information, please see chapter 5.7 User Mode Support in IM

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
default Value	false

4.6 Parameter RmEnableMultiCore

When this parameter is enabled, the RM module will adapt to run with Multicore:

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
default Value	false

4.7 Parameter RmEnableXRDCSupport

When this parameter is enabled, the Xrdc APIs are enabled to the user

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
default Value	false

4.8 Parameter RmEnableSema42Support

When this parameter is enabled, the Sema42 APIs are enabled to the user

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.9 Parameter RmMpuM7Configurable

RmMpuM7Configurable

Check this in order to be able to use the ARM Cortex M7 MPU.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.10 Parameter RmEnableMscmSupport

When this parameter is enabled, the MSCM APIs are enabled to the user

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

Property	Value
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.11 Container RmConfigSet

Vendor specific: This container is the base for a multiple configuration set

Included subcontainers:

- [Xrdc_Configuration](#)
- [Rm_XRDC_Domain_Assignment](#)
- [Rm_XRDC_Memory_Config](#)
- [Rm_XRDC_Peripheral_Config](#)
- [MPU_M7_Configuration](#)
- [Sema42_ModuleConfig](#)
- [Mscm_Configuration](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.12 Container Xrdc_Configuration

Configuration for the Xrdc.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF

Property	Value
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.13 Parameter XrdcDevErrorDetect

XrdcDevErrorDetect

Switches the Development Error Detection and Notification ON or OFF.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.14 Parameter XrdcRegistersLock

Lock XRDC registers (except PIDm registers) in order to prevent them from being unintentionally modified during Runtime.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false

Property	Value
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.15 Parameter XrdcCRLockBit

Variable for setting the CR bit lock! XrdcRegisterLock should be true

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.16 Parameter XrdcPIDRegisterLock

Select lock mode for PIDm registers

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	XRDC_PID_UNLOCKED
literals	['XRDC_PID_UNLOCKED', 'XRDC_PID_ALLOW_ACCESS_FROM_MASTER', 'XRDC_PID_FULLY_LOCKED']

4.17 Container Rm_XRDC_Domain_Assignment

All data need to configure one Domain.

Included subcontainers:

- [Domain_Master_Assignment](#)
- [Domain_Peripheral_Assignment](#)
- [Domain_Memory_Assignment](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.18 Parameter DomainID

Vendor specific:

Id for the current Domain.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	XRDC_DOMAIN0
literals	['XRDC_DOMAIN0', 'XRDC_DOMAIN1', 'XRDC_DOMAIN2', 'XRDC_DOMAIN3', 'XRDC_DOMAIN4', 'XRDC_DOMAIN5', 'XRDC_DOMAIN6', 'XRDC_DOMAIN7']

4.19 Container Domain_Master_Assignment

Vendor specific:

All data need to configure one Master.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.20 Parameter XrdcMasterInstance

Select the master of XRDC

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	Xrdc_0_CortexA53_C0
literals	['Xrdc_0_CortexA53_C0', 'Xrdc_0_CortexA53_C1', 'Xrdc_0_Debug_ETR', 'Xrdc_0_GIC_500', 'Xrdc_0_PCIE_0', 'Xrdc_0_eDMA_0', 'Xrdc_0_eDMA_1', 'Xrdc_0_M7_0_AXI', 'Xrdc_0_M7_1_AXI', 'Xrdc_0_M7_2_AXI', 'Xrdc_0_HSE', 'Xrdc_0_XRDC_GMAC_0', 'Xrdc_0_FlexRay', 'Xrdc_0_M7_0_AHB', 'Xrdc_0_M7_1_AHB', 'Xrdc_0_M7_2_AHB', 'Xrdc_0_XRDC_LLCE', 'Xrdc_0_uSDHC_Input', 'Xrdc_0_Debug_trace', 'Xrdc_1_PCIE_1', 'Xrdc_1_PFE_HIF0', 'Xrdc_1_PFE_HIF1', 'Xrdc_1_PFE_HIF2', 'Xrdc_1_PFE_HIF3', 'Xrdc_1_PFE_DDR', 'Xrdc_1_PFE_UTIL', 'Xrdc_1_USB']

4.21 Parameter XrdcPIDEnable

PID mode enable - only valid if the master instance is core type

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	XRDC_MDA_PID_DISABLE
literals	['XRDC_MDA_PID_DISABLE', 'XRDC_MDA_PID_ENABLE_EXPRESSION0', 'XRDC_MDA_PID_ENABLE_EXPRESSION1']

4.22 Parameter XrdcMasterPID

Process Identifier. This field only has meaning if the core is core master type and PID mode enable

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0
max	63
min	0

4.23 Parameter XrdcMasterPIDMask

Process Identifier Mask. This field only has meaning if the core is core master type and PID mode enable

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0
max	63
min	0

4.24 Parameter XrdcDIDBDisable

Disabled DID bypass. This field only has meaning if the core is non-core master type

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.25 Parameter XrdcMDADFMTLockBit

Variable for setting the MDA_DFMT bit lock! XrdcRegisterLock should be true.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	False

Property	Value
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.26 Parameter XrdcMasterMode

Specifies the secure attribute. This field only has meaning if the core is non-core master type

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	XRDC_SECURE
literals	['XRDC_SECURE', 'XRDC_NON_SECURE', 'XRDC_INPUT_FROM_BU←S_MASTER']

4.27 Parameter XrdcMasterPriviledgeMode

Specifies the privileged (supervisor/user) attribute. This field only has meaning if the core is non-core master type

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A

Property	Value
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	XRDC_INPUT_PRIVILEGE_FROM_BUS_MASTER
literals	['XRDC_USER_MODE', 'XRDC_PRIVILEGE_MODE', 'XRDC_INPUT← _PRIVILEGE_FROM_BUS_MASTER']

4.28 Container Domain_Peripheral_Assignment

Vendor specific:

All data need to configure Peripheral for one Domain.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.29 Parameter Secure_Access_Policy

Vendor specific:

Mode access to policy of mode Secure.

Access policy look up table:

Configuration Fields

Allowable accesses

Secure_Access_Policy

Non_Secure_Access_Policy

Secure Privileged

Secure User

NonSecure Privileged

NonSecure User

XRDC_FULL_ACCESS

XRDC_FULL_ACCESS

R,W

R,W

R,W

R,W

XRDC_FULL_ACCESS

XRDC_SUPERVISOR_ACCESS_ONLY

R,W

R,W

R,W

None

XRDC_FULL_ACCESS

XRDC_READ_ONLY

R,W

R,W

R

R

XRDC_FULL_ACCESS

XRDC_SUPERVISOR_READ_ONLY

R,W

R,W

R

None



Tresos Configuration Plug-in

XRDC_FULL_ACCESS

XRDC_NO_ACCESS

R,W

R,W

None

None

XRDC_SUPERVISOR_ACCESS_ONLY

NA

R,W

None

None

None

XRDC_READ_ONLY

NA

R

R

None

None

XRDC_NO_ACCESS

NA

None

None

None

None

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

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4.30 Parameter Non_Secure_Access_Policy

Vendor specific:

Mode access to policy of mode Non_Secure.

Access policy look up table:

Configuration Fields

Allowable accesses

Secure_Access_Policy

Non_Secure_Access_Policy

Secure Privileged

Secure User

NonSecure Privileged

NonSecure User

XRDC_FULL_ACCESS

XRDC_FULL_ACCESS

R,W

R,W

R,W

R,W

XRDC_FULL_ACCESS

XRDC_SUPERVISOR_ACCESS_ONLY

R,W

R,W

R,W

None

XRDC_FULL_ACCESS

XRDC_READ_ONLY

R,W

R,W

R

R

XRDC_FULL_ACCESS

XRDC_SUPERVISOR_READ_ONLY

R,W

R,W

R

None

XRDC_FULL_ACCESS

XRDC_NO_ACCESS

R,W

R,W

None

None

XRDC_SUPERVISOR_ACCESS_ONLY

NA

R,W

None

None

None

XRDC_READ_ONLY

NA

R

R

None

None

XRDC_NO_ACCESS

NA

None

None

None

None

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	XRDC_FULL_ACCESS
literals	['XRDC_FULL_ACCESS', 'XRDC_SUPERVISOR_ACCESS_ONLY', 'XRDC_READ_ONLY', 'XRDC_SUPERVISOR_READ_ONLY', 'XRDC_NO_ACCESS']

4.31 Reference Domain_Peripheral_Assignment

Vendor specific:

Reference to Peripheral config.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
requiresSymbolicNameValue	False
destination	/TS_T40D11M40I2R0/Rm/RmConfigSet/Rm_XRDC_Peripheral_Config

4.32 Container Domain_Memory_Assignment

Vendor specific:

All data need to configure Memory for one Domain.

Included subcontainers:

Tresos Configuration Plug-in

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.33 Parameter Secure_Access_Policy

Vendor specific:

Mode access to policy of mode Secure.

Access policy look up table:

Configuration Fields

Allowable accesses

Secure_Access_Policy

Non_Secure_Access_Policy

Secure Privileged

Secure User

NonSecure Privileged

NonSecure User

XRDC_FULL_ACCESS

XRDC_FULL_ACCESS

R,W

R,W

R,W

R,W

XRDC_FULL_ACCESS

XRDC_SUPERVISOR_ACCESS_ONLY

R,W

Tresos Configuration Plug-in

R,W

R,W

None

XRDC_FULL_ACCESS

XRDC_READ_ONLY

R,W

R,W

R

R

XRDC_FULL_ACCESS

XRDC_SUPERVISOR_READ_ONLY

R,W

R,W

R

None

XRDC_FULL_ACCESS

XRDC_NO_ACCESS

R,W

R,W

None

None

XRDC_SUPERVISOR_ACCESS_ONLY

NA

R,W

None

None

None

XRDC_READ_ONLY

NA

R

R

None

None

XRDC_NO_ACCESS

NA

None

None

None

None

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	XRDC_FULL_ACCESS
literals	['XRDC_FULL_ACCESS', 'XRDC_READ_ONLY', 'XRDC_SUPERVISOR_ACCESS_ONLY', 'XRDC_NO_ACCESS']

4.34 Parameter Non_Secure_Access_Policy

Vendor specific:

Mode access to policy of mode Non_Secure.

Access policy look up table:

Configuration Fields

Allowable accesses

Secure_Access_Policy

Non_Secure_Access_Policy

Secure Privileged

Secure User

NonSecure Privileged

NonSecure User

XRDC_FULL_ACCESS

XRDC_FULL_ACCESS

R,W

R,W

R,W

R,W

XRDC_FULL_ACCESS

XRDC_SUPERVISOR_ACCESS_ONLY

R,W

R,W

R,W

None

XRDC_FULL_ACCESS

XRDC_READ_ONLY

R,W

R,W

R

R

XRDC_FULL_ACCESS

XRDC_SUPERVISOR_READ_ONLY

R,W

R,W

R

None

XRDC_FULL_ACCESS

XRDC_NO_ACCESS

R,W

R,W

None

None

XRDC_SUPERVISOR_ACCESS_ONLY

NA

R,W

None

None

None

XRDC_READ_ONLY

NA

R

R

None

None

XRDC_NO_ACCESS

NA

None

None

None

None

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	XRDC_FULL_ACCESS
literals	['XRDC_FULL_ACCESS', 'XRDC_SUPERVISOR_ACCESS_ONLY', 'XRDC_READ_ONLY', 'XRDC_SUPERVISOR_READ_ONLY', 'XRDC_NO_ACCESS']

4.35 Reference Domain_Memory_Assignment

Choose Memory configuration

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
requiresSymbolicNameValue	False
destination	/TS_T40D11M40I2R0/Rm/RmConfigSet/Rm_XRDC_Memory_Config

4.36 Container Rm_XRDC_Memory_Config

Vendor specific:

All data need to configure Memory use for Domain.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.37 Parameter XrdcMrcInstance

Select Memory region controller will be used to control the address range below

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	Xrdc_0_DRAM0_A53
literals	['Xrdc_0_DRAM0_A53', 'Xrdc_0_DRAM0_M7_EDMA', 'Xrdc_0_DRAM0_GENERAL', 'Xrdc_0_SRAM_0', 'Xrdc_0_SRAM_1', 'Xrdc_0_SRAM_2', 'Xrdc_0_SRAM_3', 'Xrdc_0_SRAM_4', 'Xrdc_0_SRAM_5', 'Xrdc_0_SRAM_6', 'Xrdc_0_SRAM_7', 'Xrdc_0_SRAM_8', 'Xrdc_0_SRAM_9', 'Xrdc_0_SRAM_10', 'Xrdc_0_SRAM_11', 'Xrdc_0_SRAM_12', 'Xrdc_0_SRAM_13', 'Xrdc_0_SRAM_14', 'Xrdc_0_SRAM_15', 'Xrdc_0_Ncore_Regs', 'Xrdc_0_S_FLASH', 'Xrdc_0_S_LLCE', 'Xrdc_0_S_GIC_500', 'Xrdc_0_PCIE_0', 'Xrdc_0_M7_TCM_0', 'Xrdc_0_M7_TCM_1', 'Xrdc_0_M7_TCM_2', 'Xrdc_0_NoC_0', 'Xrdc_0_S_DBG_APB', 'Xrdc_1_STD_BY_SRAM', 'Xrdc_1_LLCE', 'Xrdc_1_Coherent_Accesses', 'Xrdc_1_PCIE_1', 'Xrdc_1_PFE', 'Xrdc_1_Service_Regs']

4.38 Parameter XrdcStartAddress

Start address of Memory region

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	1191182336
max	412316860416
min	0

4.39 Parameter XrdcEndAddress

End address of Memory region

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	1191706623
max	412316860416
min	0

4.40 Parameter XrdcSema4Enable

Enable Semaphore would require the domain must own the gate to have access right for this memory range

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.41 Parameter XrdcMRGDLockBit

Variable for setting the MRGD bit lock! XrdcRegisterLock should be true

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	false

4.42 Reference XrdcSema42LogicChannel

Select XrdcSema42LogicChannel to select specified Sema42 Logic Channel

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
requiresSymbolicNameValue	False
destination	/TS_T40D11M40I2R0/Rm/RmConfigSet/Sema42_ModuleConfig/Sema42↔ LogicChannelConfiguration

4.43 Container Rm_XRDC_Peripheral_Config

Vendor specific:

All data need to configure Peripheral for Domain.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.44 Parameter XrdcPeripheralSlot

Peripheral slot number of a block of peripherals. Checking Reference manual for chip specific

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	Xrdc_0_Self_Test

Property	Value
literals	['Xrdc_0_Self_Test', 'Xrdc_0_STCU2', 'Xrdc_0_MC_CGM_0', 'Xrdc_0_MC_CGM_1', 'Xrdc_0_Core_PLL', 'Xrdc_0_Peripheral_PLL', 'Xrdc_0_Accelerator_PLL', 'Xrdc_0_DRAM_PLL', 'Xrdc_0_XOSC', 'Xrdc_0_CORE_DFS', 'Xrdc_0_PERIPH_DFS', 'Xrdc_0_CMU', 'Xrdc_0_RTC', 'Xrdc_0_MC_CGM_5', 'Xrdc_0_MC_RGM', 'Xrdc_0_SRC', 'Xrdc_0_RDC', 'Xrdc_0_MC_ME', 'Xrdc_0_PMC', 'Xrdc_0_WKPU', 'Xrdc_0_SIU_L2_0', 'Xrdc_0_OCOTP', 'Xrdc_0_TMU', 'Xrdc_0_SWT_0', 'Xrdc_0_SWT_1', 'Xrdc_0_SWT_2', 'Xrdc_0_SWT_3', 'Xrdc_0_STM_0', 'Xrdc_0_STM_1', 'Xrdc_0_STM_2', 'Xrdc_0_STM_3', 'Xrdc_0_DMAMUX_0', 'Xrdc_0_DMAMUX_1', 'Xrdc_0_QuadSPI', 'Xrdc_0_DMACRC_0', 'Xrdc_0_eDMA_0', 'Xrdc_0_PIT_0', 'Xrdc_0_CRC_0', 'Xrdc_0_MSCM', 'Xrdc_0_SRAMC_0', 'Xrdc_0_SRAMC_1', 'Xrdc_0_XRDC_0', 'Xrdc_0_CAN_0', 'Xrdc_0_CAN_1', 'Xrdc_0_LINFlex_0', 'Xrdc_0_LINFlex_1', 'Xrdc_0_SPI_0', 'Xrdc_0_SPI_1', 'Xrdc_0_SPI_2', 'Xrdc_0_I2C_0', 'Xrdc_0_I2C_1', 'Xrdc_0_I2C_2', 'Xrdc_0_FTM_0', 'Xrdc_0_SAR_ADC_0', 'Xrdc_0_CU', 'Xrdc_0_SWT_4', 'Xrdc_0_SWT_5', 'Xrdc_0_SWT_6', 'Xrdc_0_SWT_7', 'Xrdc_0_HSE_MU0', 'Xrdc_0_HSE_MU1', 'Xrdc_0_HSE_MU2', 'Xrdc_0_HSE_MU3', 'Xrdc_0_STM_4', 'Xrdc_0_STM_5', 'Xrdc_0_STM_6', 'Xrdc_0_STM_7', 'Xrdc_0_DMAMUX_2', 'Xrdc_0_DMAMUX_3', 'Xrdc_0_DMACRC_1', 'Xrdc_0_eDMA_1', 'Xrdc_0_PIT_1', 'Xrdc_0_SEMA42', 'Xrdc_0_CAN_2', 'Xrdc_0_CAN_3', 'Xrdc_0_LINFlex_2', 'Xrdc_0_SPI_3', 'Xrdc_0_SPI_4', 'Xrdc_0_SPI_5', 'Xrdc_0_I2C_3', 'Xrdc_0_I2C_4', 'Xrdc_0_FTM_1', 'Xrdc_0_SAR_ADC_1', 'Xrdc_0_diPortSD_0', 'Xrdc_0_uSDHC', 'Xrdc_0_JDC', 'Xrdc_0_FR_0', 'Xrdc_0_RCCU', 'Xrdc_0_EIM', 'Xrdc_0_FCCU', 'Xrdc_0_SbSW', 'Xrdc_0_ERM_PER_ERM_EDMA0_ERM_EDMA1_ERM_7', 'Xrdc_0_ERM_CPU0_ERM_CPU1_ERM_CPU2_ERM_7', 'Xrdc_0_EIM_0_EIM_1_EIM_2_EIM_3', 'Xrdc_0_GMAC_0', 'Xrdc_0_DDRSS_0', 'Xrdc_0_PCIE_APB', 'Xrdc_1_XRDC', 'Xrdc_1_STM_TS', 'Xrdc_1_SIUL2', 'Xrdc_1_MC_CGM2', 'Xrdc_1_STDBY_SRAM_CFG', 'Xrdc_1_ERM_LLCE', 'Xrdc_1_ERM_PFE', 'Xrdc_1_ERM_Standby_RAM', 'Xrdc_1_EIM_MISC', 'Xrdc_1_EIM_LLCE', 'Xrdc_1_EIM_PFE', 'Xrdc_1_USB', 'Xrdc_1_PCIE_1']

4.45 Parameter XrdcSema4Enable

Enable Semaphore would require the domain must own the gate to have access right for this peripheral

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE

Property	Value
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.46 Parameter XrdcPDACLockBit

Variable for setting the PDAC bit lock! XrdcRegisterLock should be true.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.47 Reference XrdcSema42LogicChannel

Select XrdcSema42LogicChannel to select specified Sema42 Logic Channel

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
requiresSymbolicNameValue	False
destination	/TS_T40D11M40I2R0/Rm/RmConfigSet/Sema42_ModuleConfig/Sema42↔ LogicChannelConfiguration

4.48 Container MPU_M7_Configuration

Configuration for the ARM Cortex M7 MPU.

Included subcontainers:

- [MPU_M7_ModuleConfig](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.49 Parameter MpuDevErrorDetect

MpuDevErrorDetect

Switches the Development Error Detection and Notification ON or OFF.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.50 Container MPU_M7_ModuleConfig

Configuration for the ARM Cortex M7 MPU.

Included subcontainers:

- [RegionConfig](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.51 Parameter DefaultMapEnable

Vendor specific: Switch to indicate if the default memory map is enabled

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.52 Parameter RunInHFNMIEnable

Vendor specific: Switch to indicate if the MPU is enabled in Hard Fault or NMI

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false

Property	Value
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.53 Parameter MemManageInterruptEnable

Vendor specific: Switch to indicate if the Memory Management Interrupt is enabled

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.54 Parameter PhysicalCoreID

Physical Core ID

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	CORTEX_M7_CORE0
literals	['CORTEX_M7_CORE0', 'CORTEX_M7_CORE1', 'CORTEX_M7_CORE2']

4.55 Container RegionConfig

Vendor specific:

Configuration for interrupt requests.

Warning: This is a precompile configuration. If you uncheck a ISR, you will not be able to enable the respective channel or error functionality at post build time.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	16
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.56 Parameter RegionNumber

Vendor specific:

Region Number.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	0
max	15
min	0

4.57 Parameter StartAddress

Vendor specific:

Start Address.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	0
max	4294967294
min	0

4.58 Parameter EndAddress

Vendor specific:

End Address.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	4294967295
max	4294967295
min	32

4.59 Parameter RegionSize

Vendor specific:

RegionSize.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	1
max	9223372036854775807
min	-9223372036854775808

4.60 Parameter MemoryType

Vendor specific:

Memory type.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	MPU_M7_MEM_STRONG_ORDER
literals	['MPU_M7_MEM_STRONG_ORDER', 'MPU_M7_MEM_DEVICE_SHA← RED', 'MPU_M7_MEM_NORMAL_IO_WR_THROUGH', 'MPU_M7_M← EM_NORMAL_IO_WR_BACK1', 'MPU_M7_MEM_NORMAL_IO_NO← _CACHE', 'MPU_M7_MEM_NORMAL_IO_WR_BACK2', 'MPU_M7_M← EM_DEVICE_NOSHARE', 'MPU_M7_MEM_NORMAL_CACHEABLE']

4.61 Parameter AccessRights

Vendor specific:

Access Rights.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	MPU_M7_PRIV_UNPRIV_NONE
literals	['MPU_M7_PRIV_UNPRIV_NONE', 'MPU_M7_PRIV_RW_UNPRIV_NONE', 'MPU_M7_PRIV_RWX_UNPRIV_NONE', 'MPU_M7_PRIV_RW_UNPRIV_R', 'MPU_M7_PRIV_RWX_UNPRIV_RX', 'MPU_M7_PRIV_RW_UNPRIV_RW', 'MPU_M7_PRIV_RWX_UNPRIV_RWX', 'MPU_M7_PRIV_R_UNPRIV_NONE', 'MPU_M7_PRIV_RX_UNPRIV_NONE', 'MPU_M7_PRIV_R_UNPRIV_R', 'MPU_M7_PRIV_RX_UNPRIV_RX']

4.62 Parameter OuterCachePolicy

Vendor specific:

Outer Cache Policy.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	MPU_M7_CACHE_POLICY_NO_CACHE

Property	Value
literals	['MPU_M7_CACHE_POLICY_NO_CACHE', 'MPU_M7_CACHE_POLICY_W_BACK_WR_ALLOCATE', 'MPU_M7_CACHE_POLICY_W_THROUGH_NO_W_ALLOCATE', 'MPU_M7_CACHE_POLICY_W_BACK_NO_W_ALLOCATE']

4.63 Parameter InnerCachePolicy

Vendor specific:

Inner Cache Policy.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	MPU_M7_CACHE_POLICY_NO_CACHE
literals	['MPU_M7_CACHE_POLICY_NO_CACHE', 'MPU_M7_CACHE_POLICY_W_BACK_WR_ALLOCATE', 'MPU_M7_CACHE_POLICY_W_THROUGH_NO_W_ALLOCATE', 'MPU_M7_CACHE_POLICY_W_BACK_NO_W_ALLOCATE']

4.64 Parameter SubregionMask

Vendor specific:

Sub Region.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	0
max	255
min	0

4.65 Parameter Shareable

Vendor specific: Switch to indicate if the region is shareable

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.66 Container Sema42_ModuleConfig

Configuration for the Semaphores2.

Included subcontainers:

- [Sema42LogicChannelConfiguration](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.67 Parameter Sema42DevErrorDetect

Sema42DevErrorDetect

Switches the Development Error Detection and Notification ON or OFF.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
default Value	true

4.68 Parameter Sema42ResetAllGates

Sema42ResetAllGates

Enable or Disable calling Initialize Function

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
default Value	true

4.69 Container Sema42LogicChannelConfiguration

Configuration for Sema42 Instance

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

4.70 Parameter Sema42LogicChannel_LogicName

Vendor specific:

Logic Channel Name.

Property	Value
type	ECUC-STRING-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
default Value	SEMA42_LOGIC_CH_0

4.71 Parameter Sema42HardwareInstance

Vendor specific: Select the physical Sema42 Instance.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	true
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	SEMA42_INSTANCE0
literals	['SEMA42_INSTANCE0']

4.72 Parameter Sema42HardwareChannel

Vendor specific: Select the physical Sema42 Channel.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	0
max	15
min	0

4.73 Container Mscm_Configuration

Routing configuration for the interrupts.

Included subcontainers:

- [MscmConfig](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.74 Parameter MscmDevErrorDetect

MscmDevErrorDetect

Switches the Development Error Detection and Notification ON or OFF.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.75 Container MscmConfig

Vendor specific:

Configuration for interrupt requests.

Warning: This is a precompile configuration.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	192
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

4.76 Parameter IsrName

Vendor specific:

Interrupt Name.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	MSCM_MSCM_PCIE_1_MSI_INTERRUPT_TO_M7_A53_R52

Property	Value
literals	['MSCM_MSCM_PCIE_1_MSI_INTERRUPT_TO_M7_A53_R52', 'MSCM_MSCM_CPU_TO_M7_A53_R52_INTERRUPT_0', 'MSCM_MSCM_CPU_TO_M7_A53_R52_INTERRUPT_1', 'MSCM_MSCM_CPU_TO_M7_A53_R52_INTERRUPT_2', 'MSCM_MSCM_PCIE_0_MSI_INTERRUPT_TO_M7_A53_R52', 'MSCM_M7_CTI_INTERRUPT_0', 'MSCM_M7_CTI_INTERRUPT_1', 'MSCM_MCM', 'MSCM_EDMA_0_CH_0_15', 'MSCM_EDMA_0_CH_16_31', 'MSCM_EDMA_0_ERROR_INTERRUPT_REQUEST', 'MSCM_EDMA_1_CH_0_15', 'MSCM_EDMA_1_CH_16_31', 'MSCM_EDMA_1_ERROR_INTERRUPT_REQUEST', 'MSCM_SWT_0', 'MSCM_SWT_1', 'MSCM_SWT_2', 'MSCM_SWT_3', 'MSCM_SWT_4', 'MSCM_SWT_5', 'MSCM_SWT_6', 'MSCM_SWT_7', 'MSCM_STM_0', 'MSCM_STM_1', 'MSCM_STM_2', 'MSCM_STM_3', 'MSCM_STM_4', 'MSCM_STM_5', 'MSCM_STM_6', 'MSCM_STM_7', 'MSCM_QUADSPI_ORED_INTERRUPT_VIA_OTC', 'MSCM_QUADSPI_FLASH_A_ERROR_INTERRUPT_VIA_OTC', 'MSCM_QUADSPI_FLASH_B_ERROR_INTERRUPT_VIA_OTC', 'MSCM_STCU2', 'MSCM_USDHC', 'MSCM_FLEXCAN_0_CAN_BUS_WENT_TO_BUS_OFF_STATE', 'MSCM_FLEXCAN_0_ERROR_DETECT_ON_CAN_BUS', 'MSCM_FLEXCAN_0_MESSAGE_BUFFER_INTERRUPT_LINE_0', 'MSCM_FLEXCAN_0_LOGICAL_OR_OF_MESSAGE_BUFFER_INTERRUPT_LINES_127_TO_8', 'MSCM_FLEXCAN_1_CAN_BUS_WENT_TO_BUS_OFF_STATE', 'MSCM_FLEXCAN_1_ERROR_DETECT_ON_CAN_BUS', 'MSCM_FLEXCAN_1_MESSAGE_BUFFER_INTERRUPT_LINE_0', 'MSCM_FLEXCAN_1_LOGICAL_OR_OF_MESSAGE_BUFFER_INTERRUPT_LINES_127_TO_8', 'MSCM_FLEXCAN_2_CAN_BUS_WENT_TO_BUS_OFF_STATE', 'MSCM_FLEXCAN_2_ERROR_DETECT_ON_CAN_BUS', 'MSCM_FLEXCAN_2_MESSAGE_BUFFER_INTERRUPT_LINE_0', 'MSCM_FLEXCAN_2_LOGICAL_OR_OF_MESSAGE_BUFFER_INTERRUPT_LINES_127_TO_8', 'MSCM_FLEXCAN_3_CAN_BUS_WENT_TO_BUS_OFF_STATE', 'MSCM_FLEXCAN_3_ERROR_DETECT_ON_CAN_BUS', 'MSCM_FLEXCAN_3_MESSAGE_BUFFER_INTERRUPT_LINE_0', 'MSCM_FLEXCAN_3_LOGICAL_OR_OF_MESSAGE_BUFFER_INTERRUPT_LINES_127_TO_8', 'MSCM_PIT_0', 'MSCM_PIT_1', 'MSCM_FTM_0', 'MSCM_FTM_1', 'MSCM_GMAC_0_COMMON_INT', 'MSCM_GMAC_0_CH0_TX', 'MSCM_GMAC_0_CH0_RX', 'MSCM_GMAC_0_CH1_TX', 'MSCM_GMAC_0_CH1_RX', 'MSCM_GMAC_0_CH2_TX', 'MSCM_GMAC_0_CH2_RX', 'MSCM_GMAC_0_CH3_TX', 'MSCM_GMAC_0_CH3_RX', 'MSCM_GMAC_0_CH4_TX', 'MSCM_GMAC_0_CH4_RX', 'MSCM_SAR_ADC_0', 'MSCM_SAR_ADC_1', 'MSCM_FLEXRAY_LRAM_NON_CORRECTED_ERROR_INTERRUPT', 'MSCM_FLEXRAY_LRAM_CORRECTED_ERROR_INTERRUPT', 'MSCM_FLEXRAY_RECEIVE_FIFO_CHA_NOT_EMPTY_INTERRUPT', 'MSCM_FLEXRAY_RECEIVE_FIFO_CHB_NOT_EMPTY_INTERRUPT', 'MSCM_FLEXRAY_WAKEUP_INTERRUPT', 'MSCM_FLEXRAY_COMBINED_PROTOCOL_STATUS_AND_ERROR_INTERRUPT', 'MSCM_FLEXRAY_COMBINED_CHI_ERROR_INTERRUPT', 'MSCM_FLEXRAY_COMBINED_TRANSMIT_MESSAGE_BUFFER_INTERRUPT', 'MSCM_FLEXRAY_COMBINED_RECEIVE_MESSAGE_BUFFER_INTERRUPT', 'MSCM_FLEXRAY_COMBINED_MODULE_INTERRUPT', 'MSCM_LINFLEXD_0', 'MSCM_LINFLEXD_1', 'MSCM_LINFLEXD_2', 'MSCM_SPI_0', 'MSCM_SPI_1', 'MSCM_SPI_2', 'MSCM_SPI_3', 'MSCM_SPI_4', 'MSCM_SPI_5', 'MSCM_I2C_0', 'MSCM_I2C_1', 'MSCM_I2C_2', 'MSCM_I2C_3', 'MSCM_I2C_4', 'MSCM_MC_RGM', 'MSCM_FCCU_ALARM_STATE', 'MSCM_FCCU_MISCELLANEOUS_CONDITIONS', 'MSCM_SBSW', 'MSCM_HSE_ORED_TX_INTERRUPT_TO_MU_0', 'MSCM_HSE_ORED_RX_INTERRUPT_TO_MU_0', 'MSCM_HSE_ORED_GENERAL_PURPOSE_INTERRUPT_REQUEST',
NXP Semiconductors	S32 RM Driver

Property	Value
----------	-------

4.77 Parameter IsrTargetCore0

Vendor specific:

Select the target core for the interrupt request. Parameter is readonly if this target core is not available.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	true

4.78 Parameter IsrTargetCore1

Vendor specific:

Select the target core for the interrupt request. Parameter is readonly if this target core is not available.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	true

4.79 Parameter IsrTargetCore2

Vendor specific:

Select the target core for the interrupt request. Parameter is readonly if this target core is not available.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	true

4.80 Parameter IsrTargetCore3

Vendor specific:

Select the target core for the interrupt request. Parameter is readonly if this target core is not available.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	true

4.81 Parameter IsrTargetCore4

Vendor specific:

Select the target core for the interrupt request. Parameter is readonly if this target core is not available.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	true

4.82 Container CommonPublishedInformation

Vendor specific:

Common container, aggregated by all modules. It contains published information about vendor and versions.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.83 Parameter ArReleaseMajorVersion

Vendor specific:

Major version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF

Property	Value
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	4
max	4
min	4

4.84 Parameter ArReleaseMinorVersion

Vendor specific:

Minor version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	4
max	4
min	4

4.85 Parameter ArReleaseRevisionVersion

Vendor specific:

Revision version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	0
max	0
min	0

4.86 Parameter ModuleId

Vendor specific:

Module ID of this module from Module List.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	255
max	255
min	255

4.87 Parameter SwMajorVersion

Vendor specific:

Major version number of the vendor specific implementation of the module. The numbering is vendor specific.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	4
max	4
min	4

4.88 Parameter SwMinorVersion

Vendor specific:

Minor version number of the vendor specific implementation of the module. The numbering is vendor specific.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	0
max	0
min	0

4.89 Parameter SwPatchVersion

Vendor specific:

Patch level version number of the vendor specific implementation of the module. The numbering is vendor specific.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	2
max	2
min	2

4.90 Parameter VendorApiInfix

Vendor specific:

In driver modules which can be instantiated several times on a single ECU, BSW00347 requires

that the name of APIs is extended by the VendorId and a vendor specific name.

This parameter is used to specify the vendor specific name. In total, the

implementation specific name is generated as follows: <ModuleName>_>VendorId>_<VendorApiInfix>.

E.g. assuming that the VendorId of the implementor is 123 and the implementer chose a VendorApiInfix of "v11r456" a api name Can_Write defined in the SWS will translate to Can_123_v11r456Write.

This parameter is mandatory for all modules with upper multiplicity > 1. It shall not be used for modules with upper multiplicity =1.

Property	Value
type	ECUC-STRING-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	

4.91 Parameter VendorId

Vendor specific:

Vendor ID of the dedicated implementation of this module according to the AUTOSAR vendor list.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	43
max	43
min	43



Chapter 5

Module Index

5.1 Software Specification

Here is a list of all modules:

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Chapter 6

Module Documentation

6.1 RM Driver

6.1.1 Detailed Description

Macros

- `#define RM_STOP_SEC_CONFIG_DATA_UNSPECIFIED`

Function Reference

- void `Rm_Init` (Rm_ConfigType const *ConfigPtr)
This function initializes the RM hardware components.

6.1.2 Macro Definition Documentation

6.1.2.1 RM_STOP_SEC_CONFIG_DATA_UNSPECIFIED

```
#define RM_STOP_SEC_CONFIG_DATA_UNSPECIFIED
```

Export RM configurations.

Definition at line 116 of file CDD_Rm.h.

6.1.3 Function Reference

6.1.3.1 Rm_Init()

```
void Rm_Init (  
    Rm_ConfigType const * ConfigPtr )
```

This function initializes the RM hardware components.

This service is a non reentrant function used for driver initialization. The Initialization function shall initialize all relevant registers of the configured hardware with the values of the structure referenced by the parameter ConfigPtr. If the hardware allows for only one usage of the register, the driver module implementing that functionality is responsible for initializing the register. The initialization function of this module shall always have a pointer as a parameter, even though for Variant PC no configuration set shall be given. Instead a NULL pointer shall be passed to the initialization function.

Module Documentation

Parameters

in	<i>ConfigPtr</i>	Pointer to a selected configuration structure.
----	------------------	--

Returns

void

6.2 Mpu M7 IPV Driver

6.2.1 Detailed Description

Data Structures

- struct [Mpu_M7_Ip_RegionConfigType](#)
Configuration structure containing the region configuration. [More...](#)
- struct [Mpu_M7_Ip_ConfigType](#)
IP configuration structure. [More...](#)
- struct [Mpu_M7_Ip_ErrorDetailsType](#)
Structure used to retrieve violation details. [More...](#)

Enum Reference

- enum [Mpu_M7_Ip_CachePolicyType](#)
Enumeration listing the outer and inner cache policies when $RASR[TEX] == 0x1XX$.
- enum [Mpu_M7_Ip_MemoryType](#)
Enumeration listing the memory and cache attributes when $RASR[TEX] == 0x0XX$.
- enum [Mpu_M7_Ip_AccessRightsType](#)
Enumeration listing access permissions.
- enum [Mpu_M7_Ip_ErrorType](#)
Enumeration listing the possible access errors.

Function Reference

- void [Mpu_M7_Ip_Init_Privileged](#) (const [Mpu_M7_Ip_ConfigType](#) *pConfig)
Initializes the Memory Protection Unit general parameters and region configurations.
- void [Mpu_M7_Ip_SetRegionConfig_Privileged](#) (uint8 u8RegionNum, const [Mpu_M7_Ip_RegionConfigType](#) *const pUserConfigPtr)
Configures the region selected by u8RegionNum with the data from pUserConfigPtr.
- void [Mpu_M7_Ip_Deinit_Privileged](#) (void)
Disables the module and resets all region configurations.
- void [Mpu_M7_Ip_EnableRegion_Privileged](#) (uint8 u8RegionNum, boolean bEnable)
Enables or disabled a specific region.
- void [Mpu_M7_Ip_SetAccessRight_Privileged](#) (uint8 u8RegionNum, [Mpu_M7_Ip_AccessRightsType](#) eAccessRights)
Modify the access rights for a specific region.
- void [Mpu_M7_Ip_GetErrorRegisters](#) (uint32 *pMmfsr, uint32 *pAddress)
Retrieve fault information from SCB.

6.2.2 Data Structure Documentation

6.2.2.1 struct Mpu_M7_Ip_RegionConfigType

Configuration structure containing the region configuration.

Definition at line 145 of file [Mpu_M7_Ip_Types.h](#).

Data Fields

Type	Name	Description
uint8	u8RegionNum	Region number
uint32	u32StartAddr	Memory region start address
uint32	u32EndAddr	Memory region end address
Mpu_M7_Ip_MemoryType	eMemType	Memory Type for region
Mpu_M7_Ip_AccessRightsType	eAccessRight	Access permission for region
Mpu_M7_Ip_CachePolicyType	eOuterCachePolicy	Outer Cache Policy
Mpu_M7_Ip_CachePolicyType	eInnerCachePolicy	Inner Cache Policy
uint8	u8SubRegMask	SubRegion Disable Mask
boolean	bShareable	Enables or disables shareability

6.2.2.2 struct Mpu_M7_Ip_ConfigType

IP configuration structure.

Definition at line 170 of file Mpu_M7_Ip_Types.h.

Data Fields

Type	Name	Description
Mpu_M7_Ip_CoreType	ConfigCore	Core Id number
boolean	bDefaultMapEn	Enables the default memory map background region
boolean	bEnableRunHFNMI	Enables use of the MPU when in the hard fault, non-maskable interrupt, and FAULTMASK escalated handlers
boolean	bEnMemManageInterrupt	Enable Memory Manage Fault Exception Handler
uint8	u8RegionCnt	Region Count
const Mpu_M7_Ip_RegionConfigType *	pRegionConfigArr	Region configuration array

6.2.2.3 struct Mpu_M7_Ip_ErrorDetailsType

Structure used to retrieve violation details.

Definition at line 184 of file Mpu_M7_Ip_Types.h.

Data Fields

Type	Name	Description
uint32	u32Address	Violation address
Mpu_M7_Ip_ErrorType	eType	Type of violation

6.2.3 Enum Reference

6.2.3.1 Mpu_M7_Ip_CachePolicyType

```
enum Mpu\_M7\_Ip\_CachePolicyType
```

Enumeration listing the outer and inner cache policies when RASR[TEX] == 0x1XX.

Enumerator

MPU_M7_CACHE_POLICY_NO_CACHE	Non-cacheable
MPU_M7_CACHE_POLICY_W_BACK_WR_ALLOCATE	Write-back, write and read allocate
MPU_M7_CACHE_POLICY_W_THROUGH_NO_W_ALLOCATE	Write-through, no write allocate
MPU_M7_CACHE_POLICY_W_BACK_NO_W_ALLOCATE	Write-back, no write allocate

Definition at line 73 of file Mpu_M7_Ip_Types.h.

6.2.3.2 Mpu_M7_Ip_MemoryType

```
enum Mpu\_M7\_Ip\_MemoryType
```

Enumeration listing the memory and cache attributes when RASR[TEX] == 0x0XX.

Enumerator

MPU_M7_MEM_STRONG_ORDER	Strongly ordered
MPU_M7_MEM_DEVICE_SHARED	Shared device
MPU_M7_MEM_NORMAL_IO_WR_THROUGH	Outer and inner write-through, no write allocate
MPU_M7_MEM_NORMAL_IO_WR_BACK1	Outer and inner write-back, no write allocate
MPU_M7_MEM_NORMAL_IO_NO_CACHE	Outer and inner Non-cacheable
MPU_M7_MEM_NORMAL_IO_WR_BACK2	Outer and inner write-back; write and read allocate
MPU_M7_MEM_DEVICE_NOSHARE	Non-shared device
MPU_M7_MEM_NORMAL_CACHEABLE	Normal Memory Cache policy enabled

Definition at line 85 of file Mpu_M7_Ip_Types.h.

6.2.3.3 Mpu_M7_Ip_AccessRightsType

enum `Mpu_M7_Ip_AccessRightsType`

Enumeration listing access permissions.

Enumerator

MPU_M7_PRIV_UNPRIV_NONE	0b00000000U : — —
MPU_M7_PRIV_RW_UNPRIV_NONE	0b00000001U : rw —
MPU_M7_PRIV_RWX_UNPRIV_NONE	0b00010001U : rwx —
MPU_M7_PRIV_RW_UNPRIV_R	0b00000010U : rw r—
MPU_M7_PRIV_RWX_UNPRIV_RX	0b00010010U : rwx r-x
MPU_M7_PRIV_RW_UNPRIV_RW	0b00000011U : rw r-w-
MPU_M7_PRIV_RWX_UNPRIV_RWX	0b00010011U : rwx rwx
MPU_M7_PRIV_R_UNPRIV_NONE	0b00000101U : r —
MPU_M7_PRIV_RX_UNPRIV_NONE	0b00010101U : r-x —
MPU_M7_PRIV_R_UNPRIV_R	0b00000110U : r r—
MPU_M7_PRIV_RX_UNPRIV_RX	0b00010110U : r-x r-x

Definition at line 103 of file `Mpu_M7_Ip_Types.h`.

6.2.3.4 Mpu_M7_Ip_ErrorType

enum `Mpu_M7_Ip_ErrorType`

Enumeration listing the possible access errors.

Enumerator

MPU_M7_MEMMAN_ERROR_DATA_ACCESS	IACCVIOL Instruction access violation
MPU_M7_MEMMAN_ERROR_STACK	DACCVIOL Data access violation
MPU_M7_MEMMAN_ERROR_UNSTACK	MUNSTKERR MemManage fault on unstacking for a return from exception
MPU_M7_MEMMAN_ERROR_FLOATING_P↔OINT	MSTKERR MemManage fault on stacking for exception entry
MPU_M7_MEMMAN_ERROR_MULTIPLE	MLSPERR MemManage fault during floating point lazy state preservation
MPU_M7_MEMMAN_ERROR_NONE	MemManage fault Multiple Errors detected

Definition at line 127 of file Mpu_M7_Ip_Types.h.

6.2.4 Function Reference

6.2.4.1 Mpu_M7_Ip_Init_Privileged()

```
void Mpu_M7_Ip_Init_Privileged (
    const Mpu_M7_Ip_ConfigType * pConfig )
```

Initializes the Memory Protection Unit general parameters and region configurations.

This function is non-reentrant

Parameters

in	<i>pConfig</i>	pointer to configuration structure for MPU M7.
----	----------------	--

Returns

void

Precondition

6.2.4.2 Mpu_M7_Ip_SetRegionConfig_Privileged()

```
void Mpu_M7_Ip_SetRegionConfig_Privileged (
    uint8 u8RegionNum,
    const Mpu_M7_Ip_RegionConfigType *const pUserConfigPtr )
```

Configures the region selected by u8RegionNum with the data from pUserConfigPtr.

This function is Reentrant

Parameters

in	<i>u8RegionNum</i>	region number
in	<i>pUserConfigPtr</i>	pointer to the region configuration structure for MPU M7.

Returns

void

Precondition

6.2.4.3 Mpu_M7_Ip_Deinit_Privileged()

```
void Mpu_M7_Ip_Deinit_Privileged (
    void )
```

Disables the module and resets all region configurations.

This function is Reentrant

Returns

void

Precondition

6.2.4.4 Mpu_M7_Ip_EnableRegion_Privileged()

```
void Mpu_M7_Ip_EnableRegion_Privileged (
    uint8 u8RegionNum,
    boolean bEnable )
```

Enables or disabled a specific region.

This function is Reentrant

Parameters

in	<i>u8RegionNum</i>	: Region to be modified
in	<i>bEnable</i>	: Specifies wheter the region is enabled or disabled

Module Documentation

Returns

void

Precondition

6.2.4.5 Mpu_M7_Ip_SetAccessRight_Privileged()

```
void Mpu_M7_Ip_SetAccessRight_Privileged (
    uint8 u8RegionNum,
    Mpu_M7_Ip_AccessRightsType eRights )
```

Modify the access rights for a specific region.

This function is Reentrant

Parameters

in	<i>u8RegionNum</i>	: Region to be modified
in	<i>eRights</i>	: Specifies the new access rights

Returns

void

Precondition

6.2.4.6 Mpu_M7_Ip_GetErrorRegisters()

```
void Mpu_M7_Ip_GetErrorRegisters (
    uint32 * pMmfsr,
    uint32 * pAddress )
```

Retrieve fault information from SCB.

Parameters

out	<i>pMmfsr</i>	SCB->MMFSR register contents
out	<i>pAddress</i>	Address of the violation

Returns

void

6.3 Mscm IPV Driver

6.3.1 Detailed Description

Data Structures

- struct [Mscm_Ip_IrqRouteConfigType](#)
Structure storing the configuration for interrupt Router. [More...](#)
- struct [Mscm_Ip_ConfigType](#)
Structure storing the list of state configurations for interrupt Router. [More...](#)

6.3.2 Data Structure Documentation

6.3.2.1 struct Mscm_Ip_IrqRouteConfigType

Structure storing the configuration for interrupt Router.

Definition at line 72 of file Mscm_Ip_Types.h.

Data Fields

Type	Name	Description
uint16	u16IrqNumber	Interrupt number.
uint16	u16TargetCores	Target cores for the interrupt.

6.3.2.2 struct Mscm_Ip_ConfigType

Structure storing the list of state configurations for interrupt Router.

Definition at line 82 of file Mscm_Ip_Types.h.

Data Fields

Type	Name	Description
uint32	u32GlobalRouteConfigCount	Total of SPI interrupts routing.
const Mscm_Ip_IrqRouteConfigType *	aGlobalRouteConfig	List of interrupts routing configurations.

6.4 Sema42 IPV Driver

6.4.1 Detailed Description

Enum Reference

- enum [Sema42_Ip_StatusType](#)

Enumeration listing the possible return codes for Sema42 Ip driver.

6.4.2 Enum Reference

6.4.2.1 Sema42_Ip_StatusType

enum [Sema42_Ip_StatusType](#)

Enumeration listing the possible return codes for Sema42 Ip driver.

Enumerator

SEMA42_IP_SUCCESS	Operation successful
SEMA42_IP_ERROR	Operation resulted in error

Definition at line 87 of file Sema42_Ip_Types.h.

6.5 XRDC IPV Driver

6.5.1 Detailed Description

Data Structures

- struct [Xrdc_Ip_MemConfigType](#)
Configuration structure containing XRDC memory region configuration. [More...](#)
- struct [Xrdc_Ip_PeripheralConfigType](#)
Configuration structure containing XRDC peripheral slot configuration. [More...](#)
- struct [Xrdc_Ip_ErrorStatusType](#)
Structure used to retrieve violation details. [More...](#)
- struct [Xrdc_Ip_DomainIDErrStatusType](#)
Structure used to retrieve information violation details and the domain ID where violation occurred. [More...](#)
- struct [Xrdc_Ip_DomainConfigType](#)
Configuration structure containing XRDC domain configuration. [More...](#)
- struct [Xrdc_Ip_InstanceConfigType](#)
IP configuration structure. [More...](#)
- struct [Xrdc_Ip_ConfigType](#)
IP configuration structure. [More...](#)

Enum Reference

- enum [Xrdc_Ip_MDACInstanceType](#)
Enumeration listing MDAC instances.
- enum [Xrdc_Ip_SecureAttributeType](#)
Enumeration listing secure attributes of a XRDC master.
- enum [Xrdc_Ip_PrivilegedAttributeType](#)
Enumeration listing privilege attributes of a XRDC master.
- enum [Xrdc_Ip_PIDLockBit](#)
Enumeration listing of a PID lock registers.
- enum [Xrdc_Ip_DomainIDType](#)
Enumeration listing XRDC domain IDs.
- enum [Xrdc_Ip_MasterCoreType](#)
Enumeration listing XRDC master core.
- enum [Xrdc_Ip_MasterType](#)
Enumeration listing XRDC masters type.
- enum [Xrdc_Ip_ErrorStateType](#)
Enumeration listing states of access violations.
- enum [Xrdc_Ip_ErrorAccessType](#)
Enumeration listing errors occurred on a read or write access.
- enum [Xrdc_Ip_ErrorAttributeType](#)
Enumeration listing error attributes of access violations.

6.5.2 Data Structure Documentation

6.5.2.1 struct Xrdc_Ip_MemConfigType

Configuration structure containing XRDC memory region configuration.

Definition at line 318 of file Xrdc_Ip_Types.h.

Data Fields

Type	Name	Description
const uint32	u32XrdcMrcInstance	Corresponding MRC instance of current memory region.
const uint32	u32XrdcMrcRegionDescriptor	Selection of descriptor for current memory region.
const uint32	u32XrdcStartAddress	Start address of current memory region.
const uint32	u32XrdcEndAddress	End address of current memory region.
const uint32	u32XrdcSema4Enable	Enable or disable Semaphore support.
const uint32	u32XrdcSema4Number	Semaphore number used in access evaluation.
const uint32	u32XrdcMRGDLockBit	Enable or disable MRGD bit lock.
const uint32	u32XrdcMemPolicy	Access policy of current memory region.
const uint32	u32XrdcMemPolicy1	

6.5.2.2 struct Xrdc_Ip_PeripheralConfigType

Configuration structure containing XRDC peripheral slot configuration.

Definition at line 334 of file Xrdc_Ip_Types.h.

Data Fields

Type	Name	Description
const uint32	u32XrdcPdacfInstance	Corresponding PDAC instance of current peripheral slot.
const uint32	u32XrdcSema4Enable	Enable or disable Semaphore support.
const uint32	u32XrdcSema4Number	Semaphore number used in access evaluation.
const uint32	u32XrdcPDACLockBit	Enable or disable PDAC bit lock.
const uint32	u32XrdcPerPolicy	Access policy of current peripheral slot.
const uint32	u32XrdcPerPolicy1	

6.5.2.3 struct Xrdc_Ip_ErrorStatusType

Structure used to retrieve violation details.

Definition at line 348 of file Xrdc_Ip_Types.h.

Data Fields

Type	Name	Description
uint32	u32AddError	Address of an access violation.
uint32	u32AddErrorRemain	
Xrdc_Ip_ErrorStateType	ErrState	State of access violations.
uint32	u32ErrPort	Port number of the MRC that detected the access violation.
Xrdc_Ip_ErrorAccessType	ErrAccess	Whether the captured access violation occurred on a read or write access.
Xrdc_Ip_ErrorAttributeType	ErrAttribute	Attributes of the access violation.

6.5.2.4 struct Xrdc_Ip_DomainIDErrStatusType

Structure used to retrieve information violation details and the domain ID where violation occurred.

Definition at line 362 of file Xrdc_Ip_Types.h.

Data Fields

Type	Name	Description
Xrdc_Ip_DomainIDType	DomainIDAccessError	Domain ID of the access violation.
Xrdc_Ip_ErrorStatusType	ErrorStatus[XRDC_MAX_OF_ERROR_RECORDS]	Attributes of the access violation.

6.5.2.5 struct Xrdc_Ip_DomainConfigType

Configuration structure containing XRDC domain configuration.

Definition at line 371 of file Xrdc_Ip_Types.h.

Data Fields

Type	Name	Description
Xrdc_Ip_DomainIDType	u32XrdcDomainID	XRDC domain ID.
Xrdc_Ip_MDACInstanceType	XrdcMdacInstance	Corresponding MDA instance of current master.
Xrdc_Ip_MasterType	XrdcCoreMdacInstance	Core or noncore attribute of current master.
const uint32	u32XrdcProcessID	PID value of curent domain ID.
const uint32	u32XrdcProcessIDMask	PIDM value of current domain ID.
const uint32	u32XrdcProcessIDEnable	Enable or disable PID support.
const uint32	u32XrdcThreeStateModel	If core master support three-state model or not.
const uint32	u32XrdcWordDescriptor	Selection of descriptor for current core master.

Data Fields

Type	Name	Description
const uint32	u32XrdcDIDBypassDisable	Enable or disable DID bypass.
const uint32	u32XrdcMDADFMTLockBit	Enable or disable MDA_DFMT bit lock.
Xrdc_Ip_SecureAttributeType	eXrdcSecureOutput	Secure attribute of current master.
Xrdc_Ip_PrivilegedAttributeType	eXrdcPrivilegedOutput	Privileged attribute of current master.

6.5.2.6 struct Xrdc_Ip_InstanceConfigType

IP configuration structure.

Definition at line 393 of file Xrdc_Ip_Types.h.

Data Fields

Type	Name	Description
const uint32	u32XrdcInstance	XRDC instance.
const Xrdc_Ip_DomainConfigType *	pDomainConfig	Pointer to a list of configured domains.
const uint32	u32DomainConfigCnt	Number of configured domains.
const Xrdc_Ip_MemConfigType *	pMemoryDesConfig	Pointer to a list of configured memory regions.
const uint32	u32MemoryConfigCnt	Number of configured memory regions.
const Xrdc_Ip_PeripheralConfigType *	pPeripheralDescriptorConfig	Pointer to a list of configured peripheral slots.
const uint32	u32PeripheralConfigCnt	Number of configured peripheral slots.
const uint32	u32XrdcCRLockBit	Enable or disable CR bit lock.
Xrdc_Ip_PIDLockBit	XrdcPIDLockBit	Enable or disable CR bit lock.

6.5.2.7 struct Xrdc_Ip_ConfigType

IP configuration structure.

Definition at line 409 of file Xrdc_Ip_Types.h.

Data Fields

Type	Name	Description
const Xrdc_Ip_InstanceConfigType * const *const	pInstanceConfig	Pointer to a list of configured instances.
const uint32	u32InstanceConfigCnt	Number of configured instances.
const uint32 *const	pInstanceInUsed	Pointer to a list of configured XRDC instances.
const uint32	u32InstanceInUsedCnt	Number of configured XRDC instances.

6.5.3 Enum Reference

6.5.3.1 Xrdc_Ip_MDACInstanceType

enum `Xrdc_Ip_MDACInstanceType`

Enumeration listing MDAC instances.

Definition at line 148 of file `Xrdc_Ip_Types.h`.

6.5.3.2 Xrdc_Ip_SecureAttributeType

enum `Xrdc_Ip_SecureAttributeType`

Enumeration listing secure attributes of a XRDC master.

Definition at line 194 of file `Xrdc_Ip_Types.h`.

6.5.3.3 Xrdc_Ip_PrivilegedAttributeType

enum `Xrdc_Ip_PrivilegedAttributeType`

Enumeration listing privilege attributes of a XRDC master.

Definition at line 204 of file `Xrdc_Ip_Types.h`.

6.5.3.4 Xrdc_Ip_PIDLockBit

enum `Xrdc_Ip_PIDLockBit`

Enumeration listing of a PID lock registers.

Definition at line 214 of file `Xrdc_Ip_Types.h`.

6.5.3.5 Xrdc_Ip_DomainIDType

enum `Xrdc_Ip_DomainIDType`

Enumeration listing XRDC domain IDs.

Definition at line 224 of file `Xrdc_Ip_Types.h`.

6.5.3.6 Xrdc_Ip_MasterCoreType

enum `Xrdc_Ip_MasterCoreType`

Enumeration listing XRDC master core.

most significant byte is XRDC instance. least significant byte is Master core number. Ex: 0x110 - XRDC instance 1, Master core number 16.

Definition at line 251 of file `Xrdc_Ip_Types.h`.

6.5.3.7 Xrdc_Ip_MasterType

enum `Xrdc_Ip_MasterType`

Enumeration listing XRDC masters type.

Definition at line 267 of file `Xrdc_Ip_Types.h`.

6.5.3.8 Xrdc_Ip_ErrorStateType

enum `Xrdc_Ip_ErrorStateType`

Enumeration listing states of access violations.

Definition at line 278 of file `Xrdc_Ip_Types.h`.

6.5.3.9 Xrdc_Ip_ErrorAccessType

enum `Xrdc_Ip_ErrorAccessType`

Enumeration listing errors occurred on a read or write access.

Definition at line 289 of file `Xrdc_Ip_Types.h`.

6.5.3.10 Xrdc_Ip_ErrorAttributeType

enum `Xrdc_Ip_ErrorAttributeType`

Enumeration listing error attributes of access violations.

Definition at line 300 of file `Xrdc_Ip_Types.h`.

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